

Satellite Communications

ECE 4644 / 5610 Spring 2026

Week #4

Feb 9 - 13, 2026

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Topics (1/2)

- Review of Kepler's Laws and orbital elements
- Sidereal day versus Solar day
- Geosynchronous orbit, Geostationary orbit
- Orbital perturbations
- Precession
- Sun-synchronous orbit

Topics (2/2)

- Orbital effects
- Doppler frequency shift
- Eclipse effects
- Launching satellites
- Satellite systems

Review – Kepler's laws and Orbital elements

- Kepler's three laws describe the orbit of a satellite
- The solution for the equation of the orbit is obtained by solving Newton's expression, $\mathbf{F} = m\mathbf{a}$, for a satellite subject to gravity
- Specification of a satellite orbit in orbital plane coordinates requires three parameters, (a, e, t_p)
- Specification of a satellite orbit with respect to Earth requires six parameters - the orbital elements symbolized by $(a, e, t_p, \Omega, \omega, i)$

Sidereal versus Solar Day: Review

- Earth makes one complete rotation with respect to the distant stars in one *sidereal day*
- One sidereal day = 23 h 56 min 4.091 s
- Earth makes one complete rotation with respect to the sun in one *solar day*
- One solar day = 24 h
- The difference of 3 min 56 s is due to Earth's movement in its orbit around the sun

Geosynchronous Orbit

- An orbit about Earth with a period of one sidereal day ($23^{\text{hr}} 56^{\text{mn}} 4^{\text{sc}}$) is called a *geosynchronous* orbit (GSO)
- A satellite in GSO returns to exactly the same position in the sky after one sidereal day (for an Earth-based observer)
- If a GSO is circular ($e = 0$) and also lies in the equatorial plane ($i = 0$) then it is also a *geostationary* orbit (GEO)

Orbital Perturbations

- Preceding theory assumed that only Earth's gravitational force affects the satellite
- A satellite in orbit is weightless
- Any force, however small, will cause acceleration (à la Newton)
- The acceleration will change the orbit
- There are many forces acting on a satellite in orbit

Orbital Perturbations

- **Moon's gravity**
- **Sun's gravity**
- Earth's tides
- Earth's magnetic field
- **Solar wind**
- Meteorite hits
- **Non-spherical earth**
- **Atmosphere (LEO only)**

Note: Influence of moon dominates all other perturbing forces once satellite is above Earth's atmosphere

Orbital Perturbations

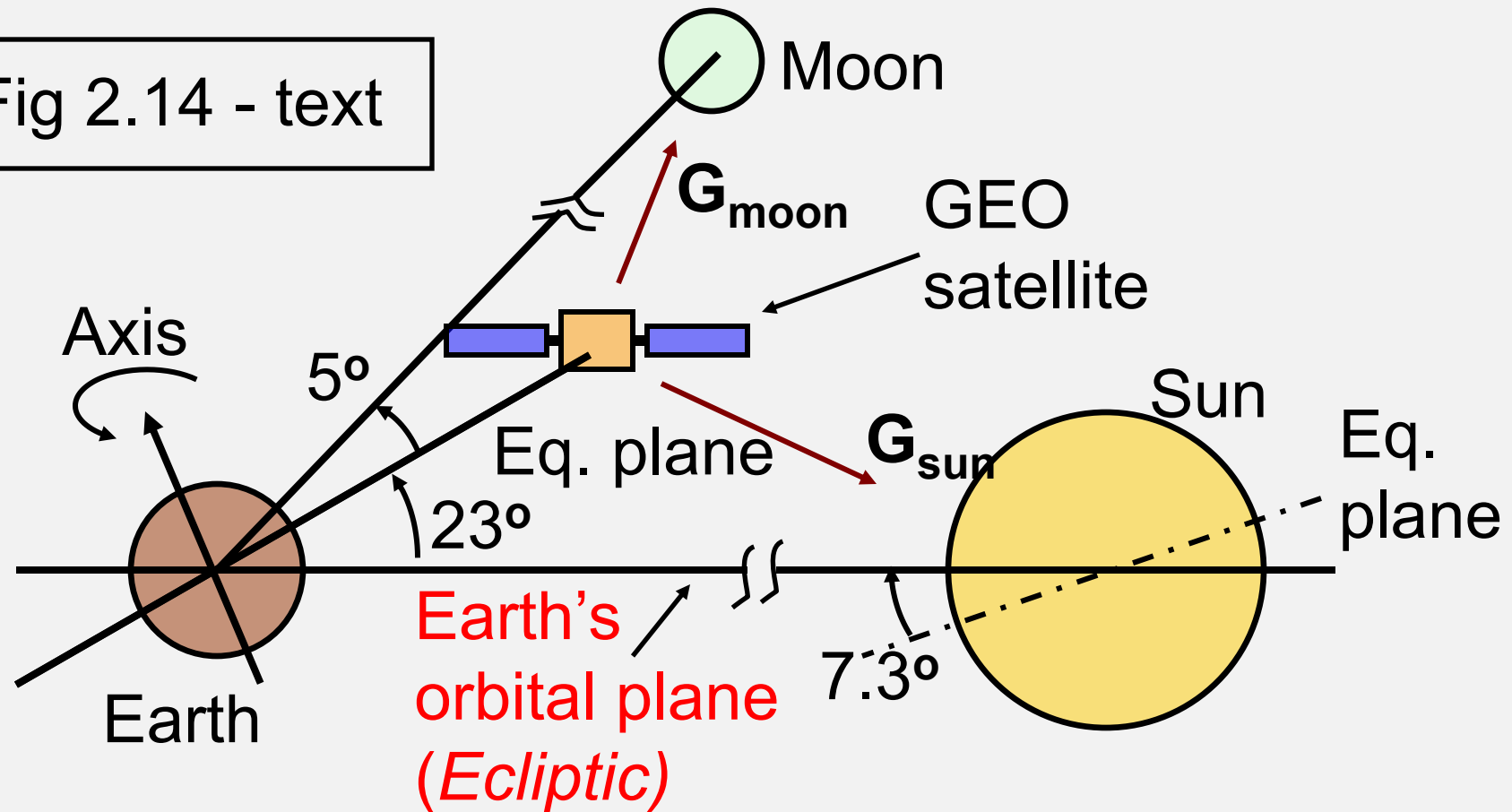
- At LEO orbit the primary orbit perturbation is due to *atmospheric drag*
- At GEO orbit the primary perturbations are due to gravitational pulls of the Moon and Sun

Relative to the orbital plane of the satellite:

- *Out-of-plane* forces cause the orbit inclination, i , to change
- *In-plane* forces affect orbit longitude factors (e, t_p, ω)

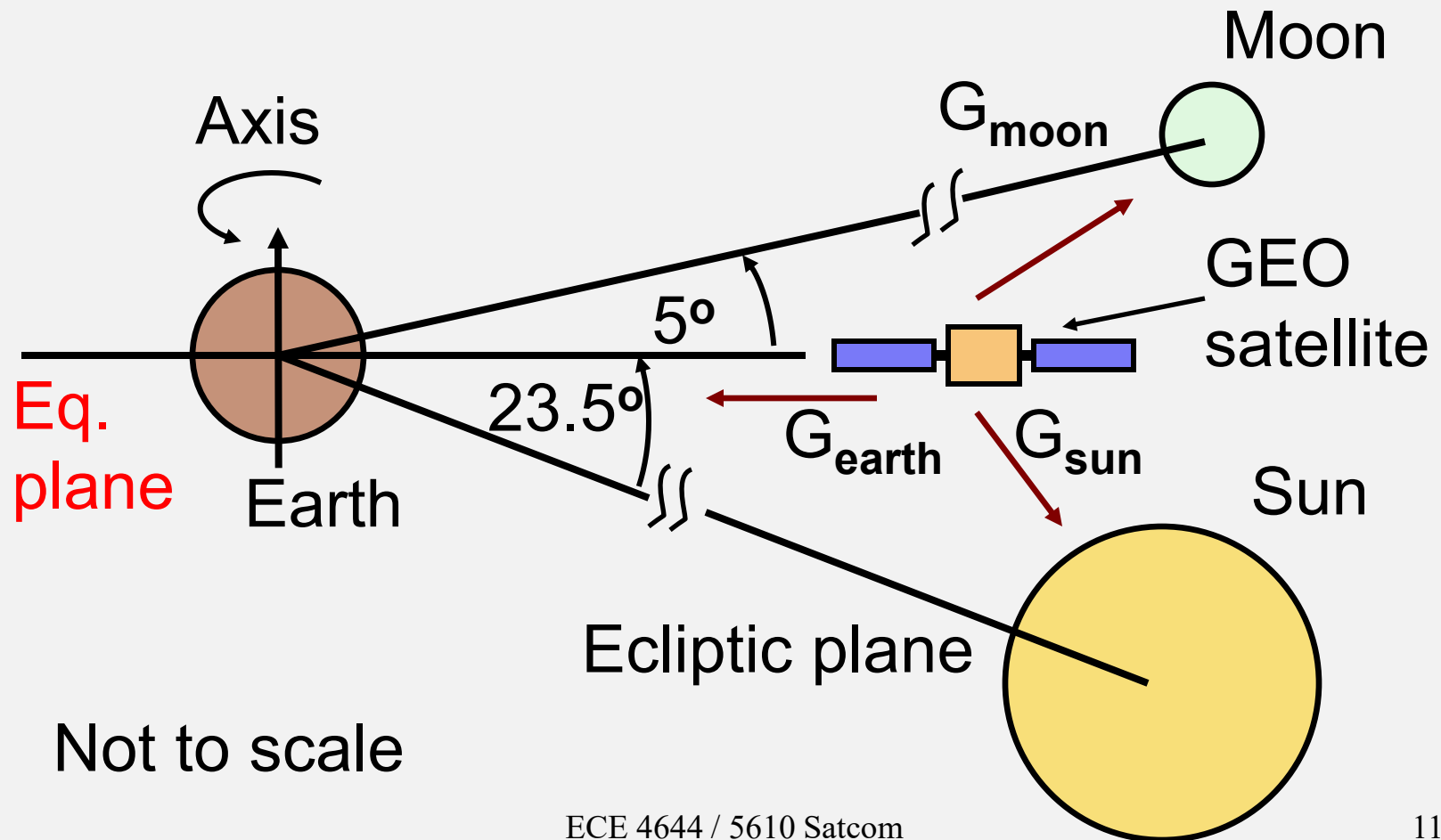
Orbital Perturbations: Effect of Moon and Sun

Fig 2.14 - text



The sun, the earth, and the moon do not have common orbital planes

Fig 2.14 - redrawn



Orbital Perturbations: Moon and Sun

- Geometry of Sun-Earth-Moon
 - The *Ecliptic* (Earth's orbital plane) is inclined at 7.3° to Sun's equator
 - Earth's axis is tilted 23.5° away from the normal to the ecliptic
 - The moon's orbit around the earth is tilted 5° away from Earth's equatorial plane
- The resulting out-of-plane forces change the inclination of a satellite orbit
- GEO satellite: i changes by $\sim 0.85^\circ$ /year

Orbital Perturbations: In-plane forces

- Longitudinal Drift
 - Earth is not a perfect sphere (equatorial bulge, Mascons)
 - Varying gravitational forces in the plane of orbit cause satellite to drift in longitude
 - There are four equilibrium points in GEO
 - Two stable (wells), two unstable (hills)
 - Satellite tends to drift towards stable points

Orbital Perturbations

- Longitudinal Drift

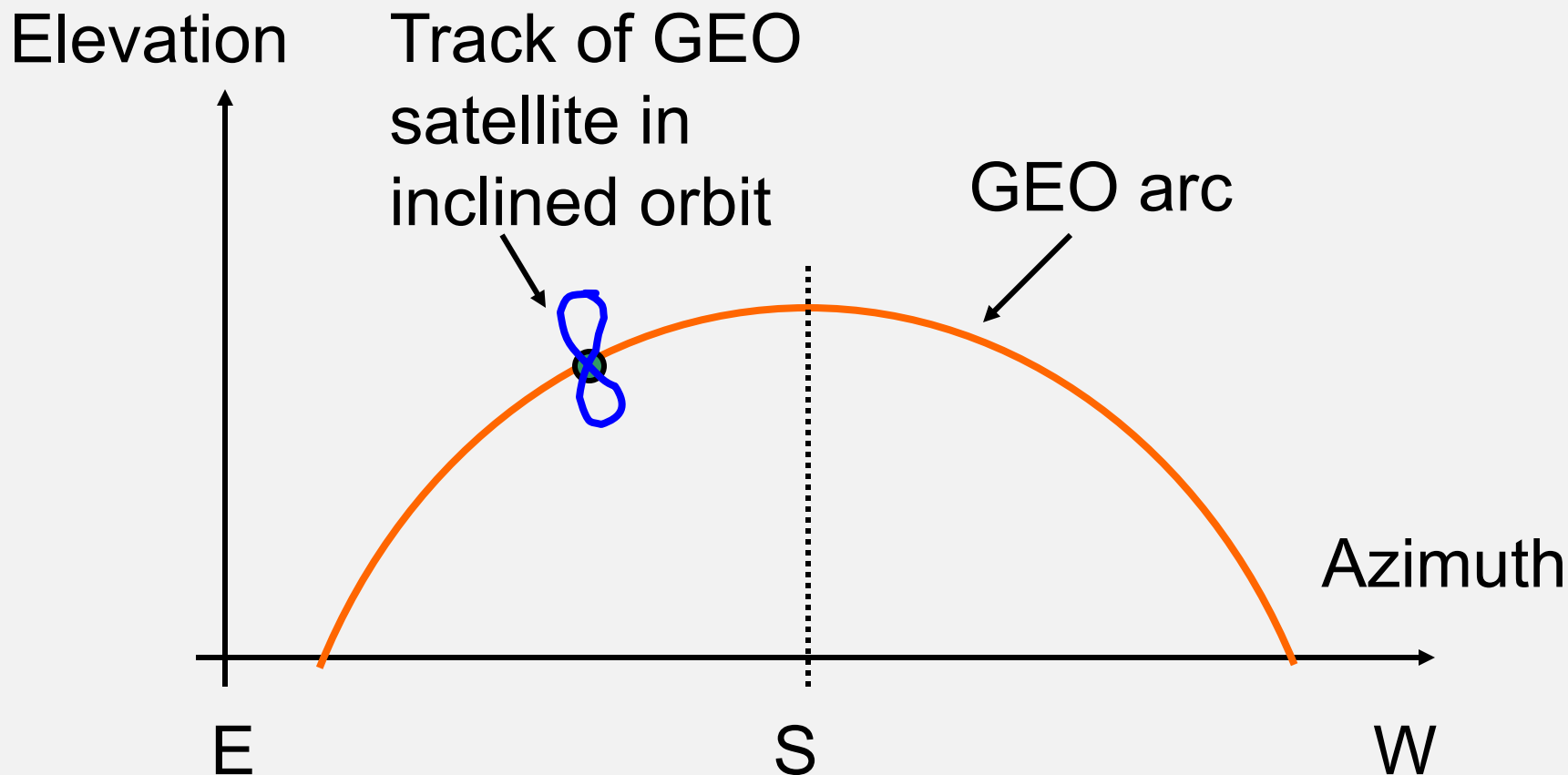
Stable points: 75° and 252° E

Unstable points: 162° and 348° E

- Satellite must be given a velocity increment to correct drift from time to time (station keeping)
- Stable points are called *graveyard geosynchronous orbit locations* (not to be confused with *graveyard orbits*)

Orbital Perturbations: GEO orbit

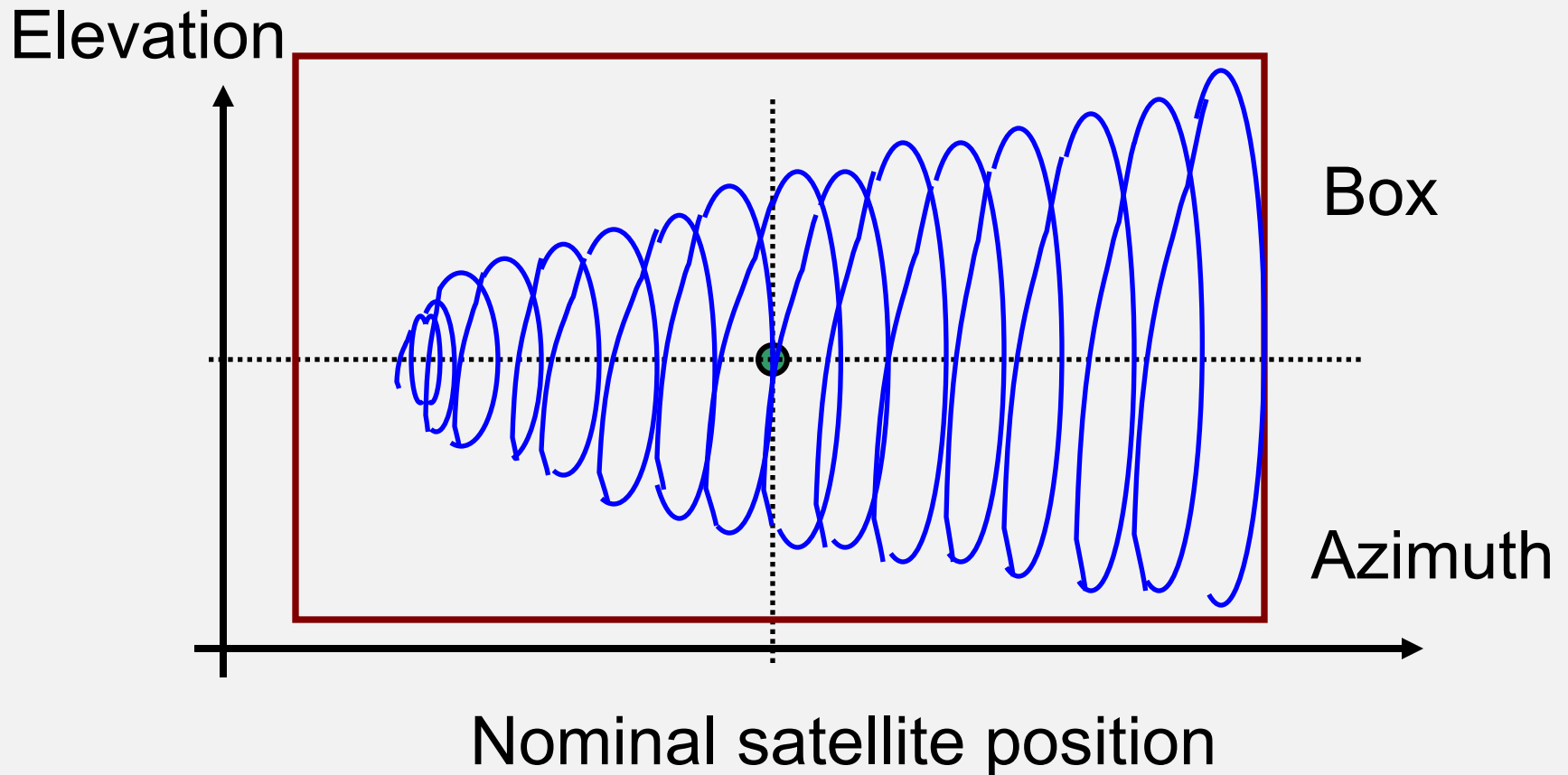
- Inclination changes slowly
- GEO orbit becomes inclined to earth's equatorial plane
- Sub-satellite point moves in latitude
- Satellite appears to move normal to GEO arc in a figure-of-eight over a sidereal day
- Peak angle variation normal to GEO arc is approximately equal to inclination angle



A GEO satellite in a stable inclined orbit appears to execute a figure-of-eight track every 24 hours

GEO Satellite Tracks

- Combined effect of changing inclination and E or W drift gives looping track in sky
- Usually satellite must be maintained within an Az-El box in sky - typical GEO box is $\pm 0.05^\circ$
- Rocket motors on satellite must be fired at intervals to correct drifts



Track of GEO satellite in an orbit with changing inclination and longitudinal drift. Each loop is 24 hours

Station Keeping Maneuvers

- Satellite orbit corrections are called station keeping maneuvers
- Rocket motors on satellite are fired every four to six weeks
- Inclination is set to small negative value by rocket burn normal to orbit, e.g., -0.04°
- Inclination changes about 0.014° per week
- After six weeks, inclination is $+0.044^\circ$

Station Keeping Maneuvers

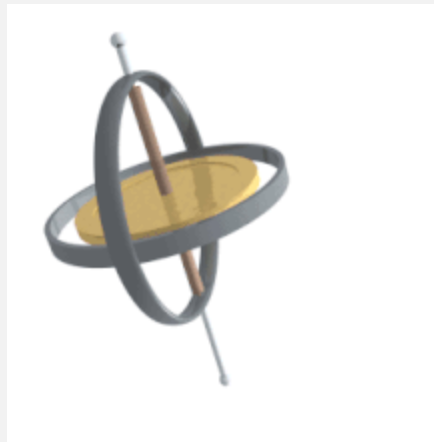
- Longitudinal drift is corrected by increasing (or decreasing) satellite velocity
- Satellite is pushed past its nominal position
- Then drifts back slowly over next six weeks
- Station keeping requires a reliable in-orbit propulsion system

Station Keeping Maneuvers

- N-S station keeping (inclination) requires ten times as much energy (hence fuel) as E-W station keeping (longitude)
- Station keeping fuel may be 50% of initial on-orbit weight of GEO satellite with 15 year life
- Satellites increasingly use arc jets and ion thrusters to reduce weight of fuel
- Electric propulsion is increasingly preferred over chemical propulsion

Orbital Perturbations: Precession Effects

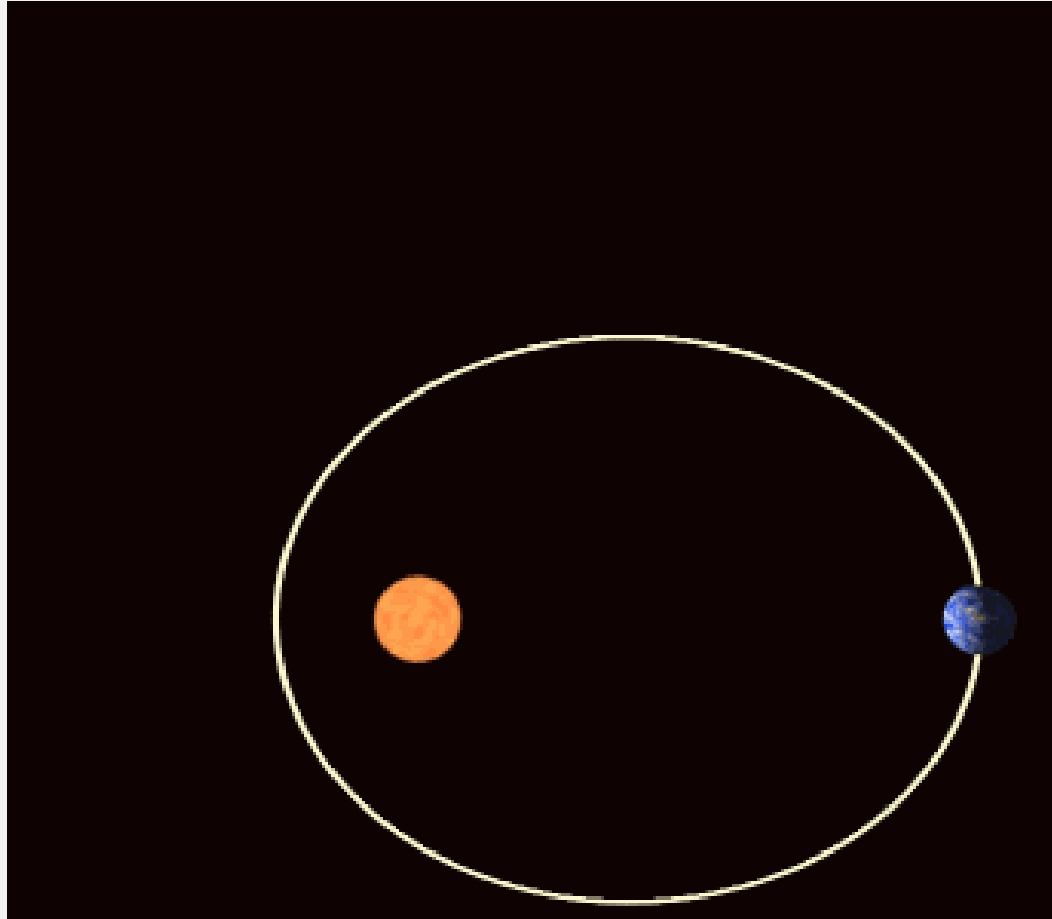
- The equatorial bulge has a gravitational effect that causes an orbit to *precess* around the rotational axis of Earth
- Classic example of precession: A spinning top with an axis of rotation that itself rotates about an axis



Orbital Perturbations: Precession Effects

- Two types of orbital precession:
- *Rotation of the line of apsides:* the positions of perigee and apogee shift around *in the orbital plane*, i.e., the argument of perigee, ω , changes with time
- *Regression of the nodes:* the positions of the nodes *in the equatorial plane* shift around in longitude, i.e., the RA of the ascending node, Ω , changes with time

Orbital Precession: Rotation of the line of apsides



By WillowW - Own work, CC BY 3.0, <https://commons.wikimedia.org/w/index.php?curid=3416065>

Inertial Versus Sun-Synchronous orbit

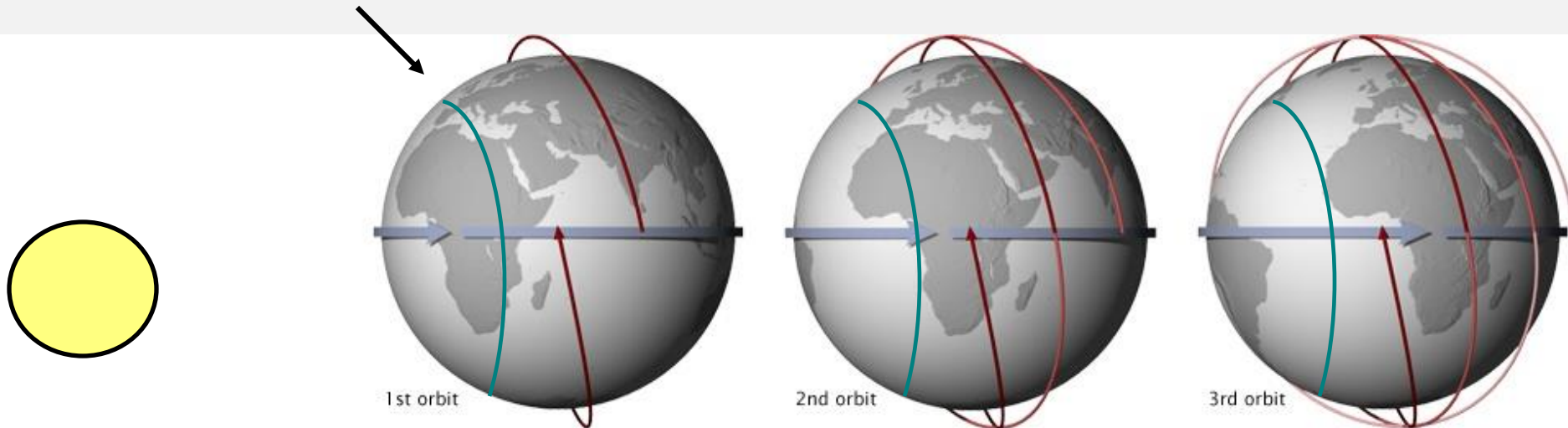
- By default the orbital plane of a satellite has an orientation that is fixed in the inertial reference frame (wrt the distant stars)
- Consider: a satellite in a polar orbit will move in the noon-midnight plane of Earth at one time and then in the dusk-dawn plane at another as Earth revolves around the Sun
- It might be useful to have the orbit fixed to one of these planes, or to any particular Local Time (LT) plane

Sun-Synchronous orbit

- An orbit that maintains a constant local time along its path is called *sun-synchronous*
- The Earth rotates under the sun-synchronous orbit which is fixed relative to the Sun
- To achieve this, the orbital nodes must shift position *eastward* at a rate of 0.9856° per day
- That is, we require regression of the nodes
- Can be done with a suitable combination of a , e , and i

Sun Synchronous orbit

Noon meridian



Sun

The satellite orbit is aligned with $\sim 1:30$ LT

Real versus Keplerian Orbits (1/2)

- Because of the various perturbation effects, perfectly constant, elliptical *Keplerian* orbits are not realized
- Since the real orbit is not a closed path like an ellipse, orbital period is not well-defined
- The *anomalistic period*, the time between perigee passes, is often specified instead

Real versus Keplerian Orbits (2/2)

- Modeling 'real' orbits starts with an *osculating orbit*, i.e., a Keplerian orbit described by $(a_o, e_o, t_p, \Omega_o, \omega_o, i_o)$ at an instant of time t_o
- At time t , apply the time derivative terms, da/dt , de/dt , etc., and the difference $t-t_o$, to generate revised values of a , e , etc.
- The NASA two-line orbital bulletins contain this information (*secular variation*)

Other Orbital Effects

- Doppler frequency shift:
- Radiowave signal received by a stationary observer from a transmitter on a moving platform appears to be at an increased or decreased frequency
- Movement towards observer increases frequency – and vice-versa
- A significant factor in LEO satcom design

Doppler Shift

Motion towards (or away from) an observer will cause an increase (or decrease) in observed frequency

$$f_{\text{doppler}} = V_T / \lambda$$

E.g. Shuttle, $V_T = 1585 \text{ m/s}$

Frequency = 2300 MHz

Doppler shift is 12.2 kHz max...

...and varies with time between -12.2 kHz and 12.2 kHz

V_T = radial
velocity towards
observer

Eqn. 2.44 in text

Solar Eclipse

- Earth gets between the sun and the satellite causing darkness
- For GEO systems, eclipses occur in Spring and Fall when the sun appears on or near the celestial equator (in the sky)
- For LEO systems, conditions for eclipse vary with satellite orbit inclination, occurs once every orbit for almost half period

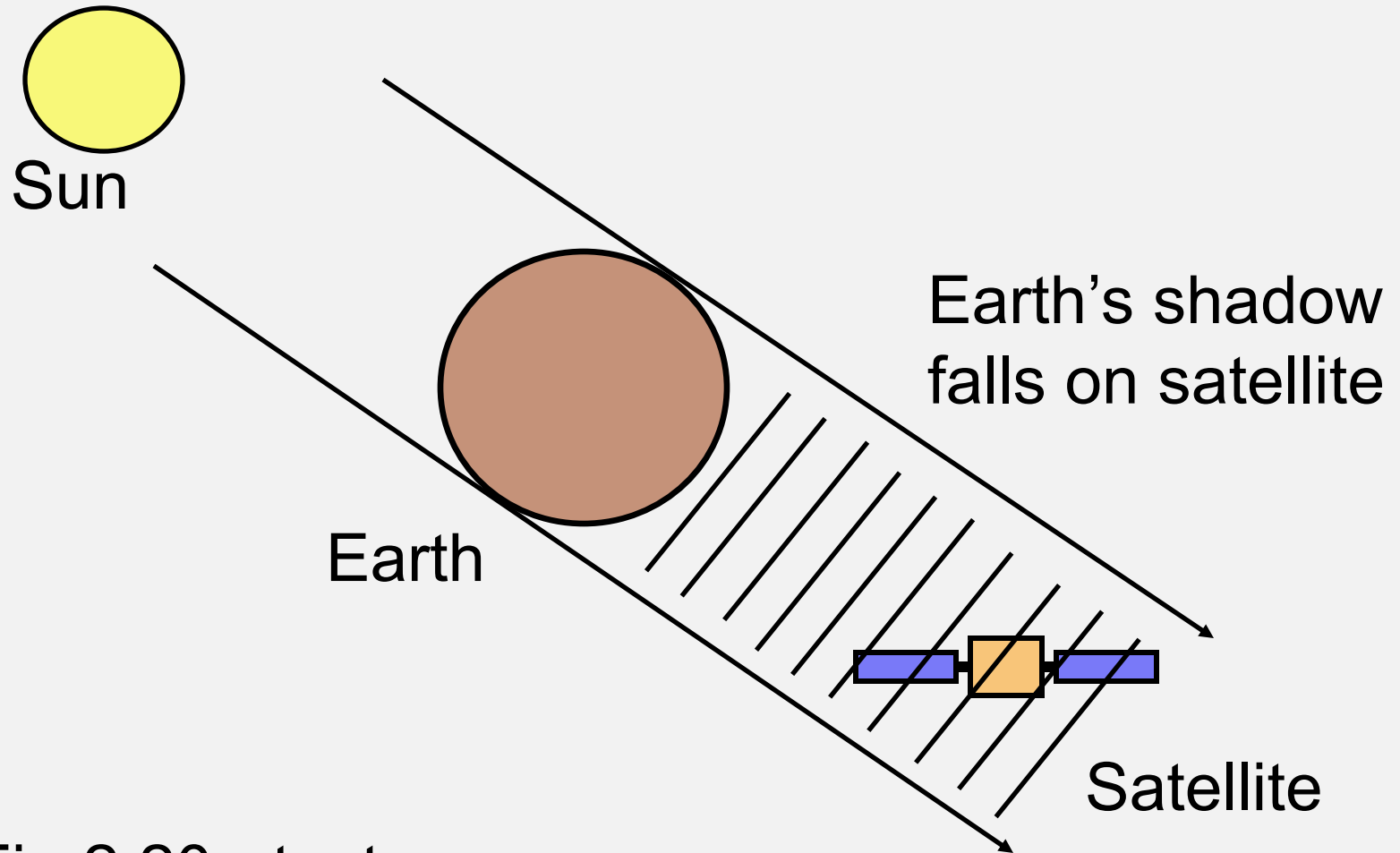


Fig 2.20 - text
Geometry of solar eclipse

Solar Eclipse

- The source of power for satellite operation is solar panels
- Operation during an eclipse requires batteries that store solar energy
- Note that temperatures and thermal balance change significantly and rapidly through an eclipse, leading to heat stresses

Solar Eclipse

- Eclipses occur for GEO satellites in Spring and Fall
- Maximum duration is 70 minutes around the equinoxes – Mar 23, Sept 23
- Satellite must have sufficient battery capacity to operate at full power for > 70 minutes
- Battery capacity is rated in ampere-hours

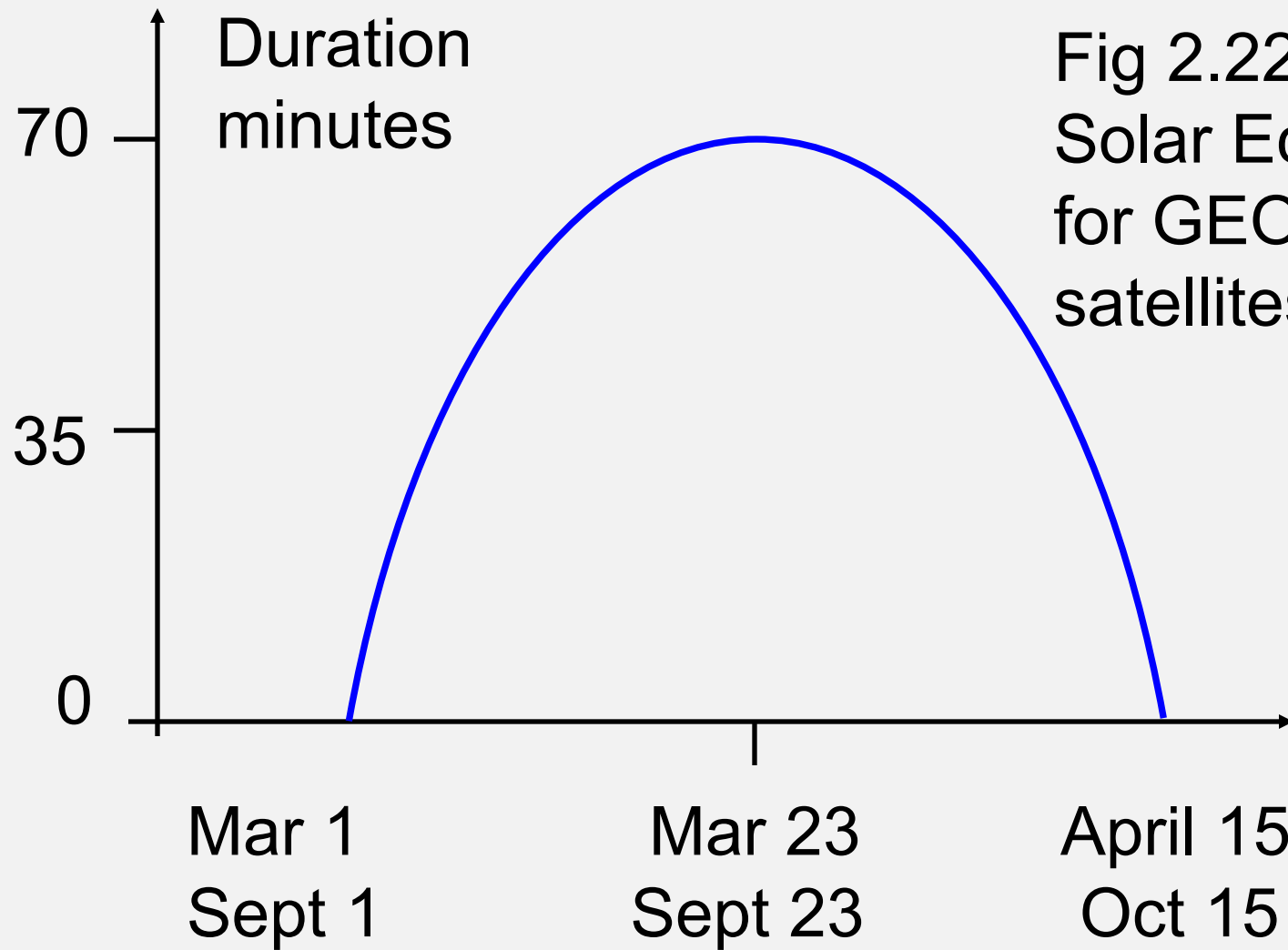


Fig 2.22 - text
Solar Eclipse
for GEO
satellites

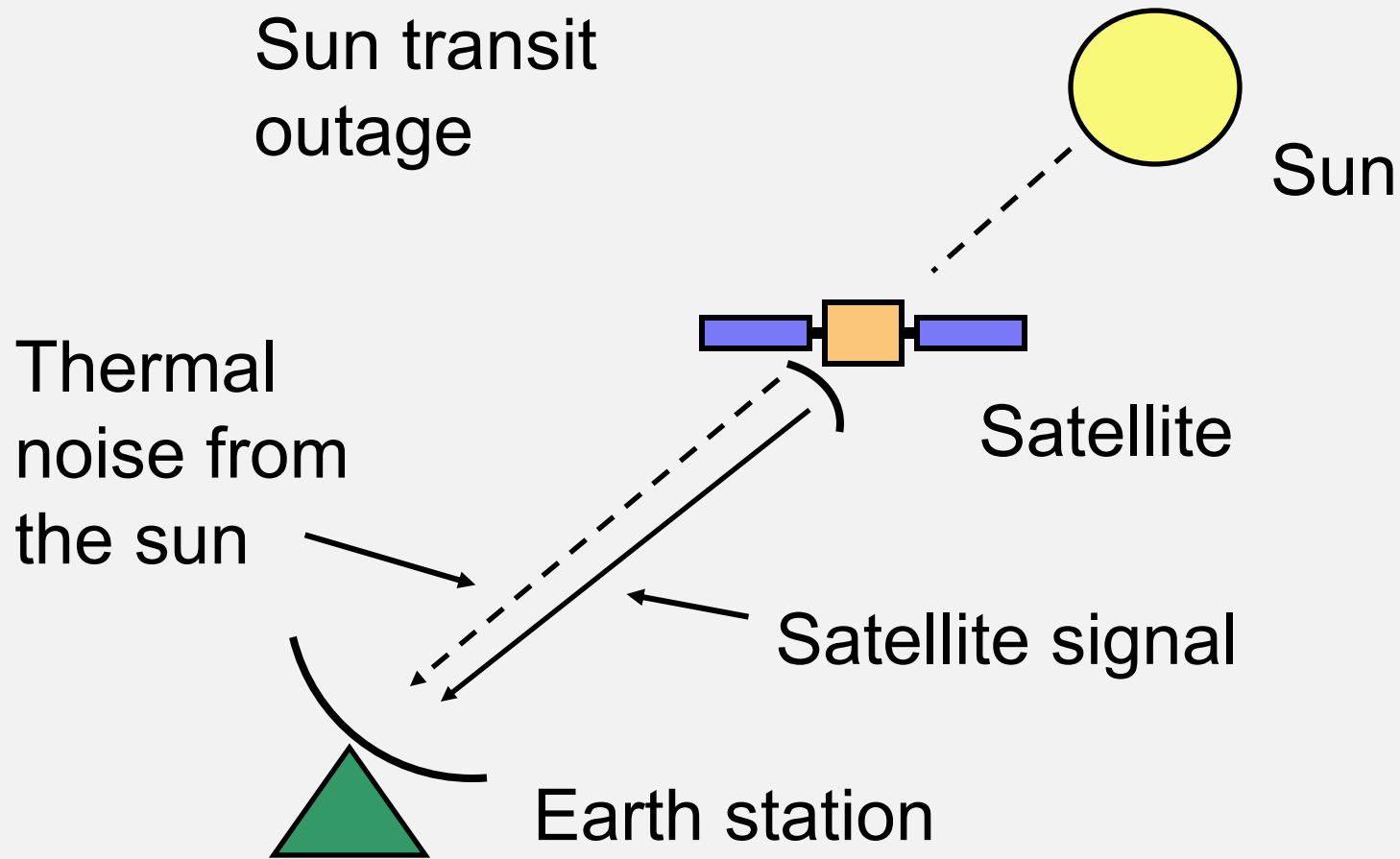
Satellite Batteries

- High capacity, high reliability, metal hydride
- Charged from solar cells via controller
- Typical bus voltage is 50 volts
- For 6 kW satellite current drain is 120 A
- 120 A for 70 minutes = 140 AH
- Battery capacity will be ~ 250 AH
- Batteries must not discharge 100%

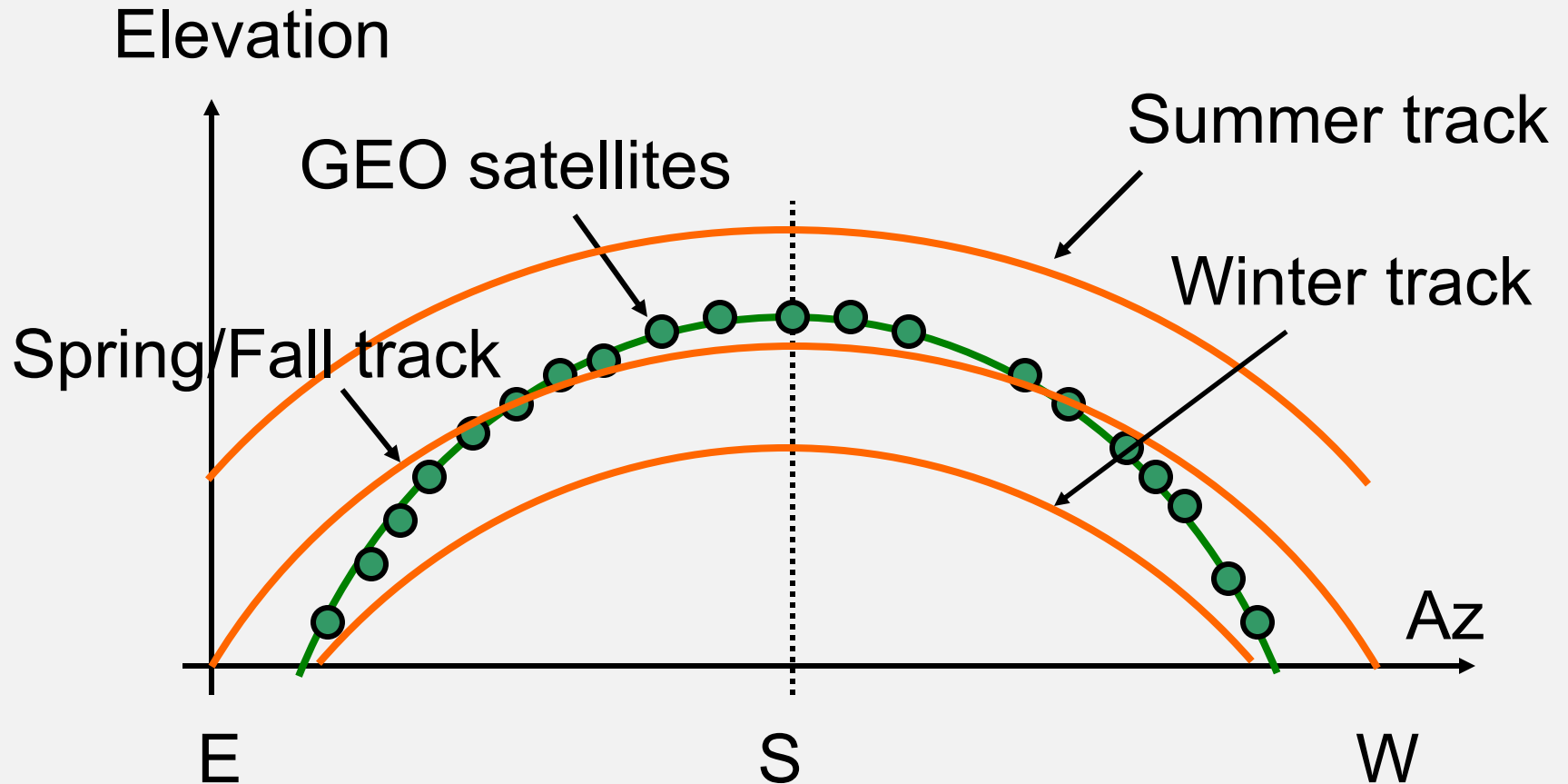
Sun Transit Outage

- During equinox periods satellite also passes in front of sun
- Sun is a powerful source of microwave noise
- Surface temperature at microwave frequencies is 6000 – 10,000 K
- Noise strength can exceed signal strength
- Earth stations with narrow beam antennas may suffer communications failure for a few minutes

Fig 2.22 - text
Sun transit
outage



Daily Track of Sun in Sky by Season



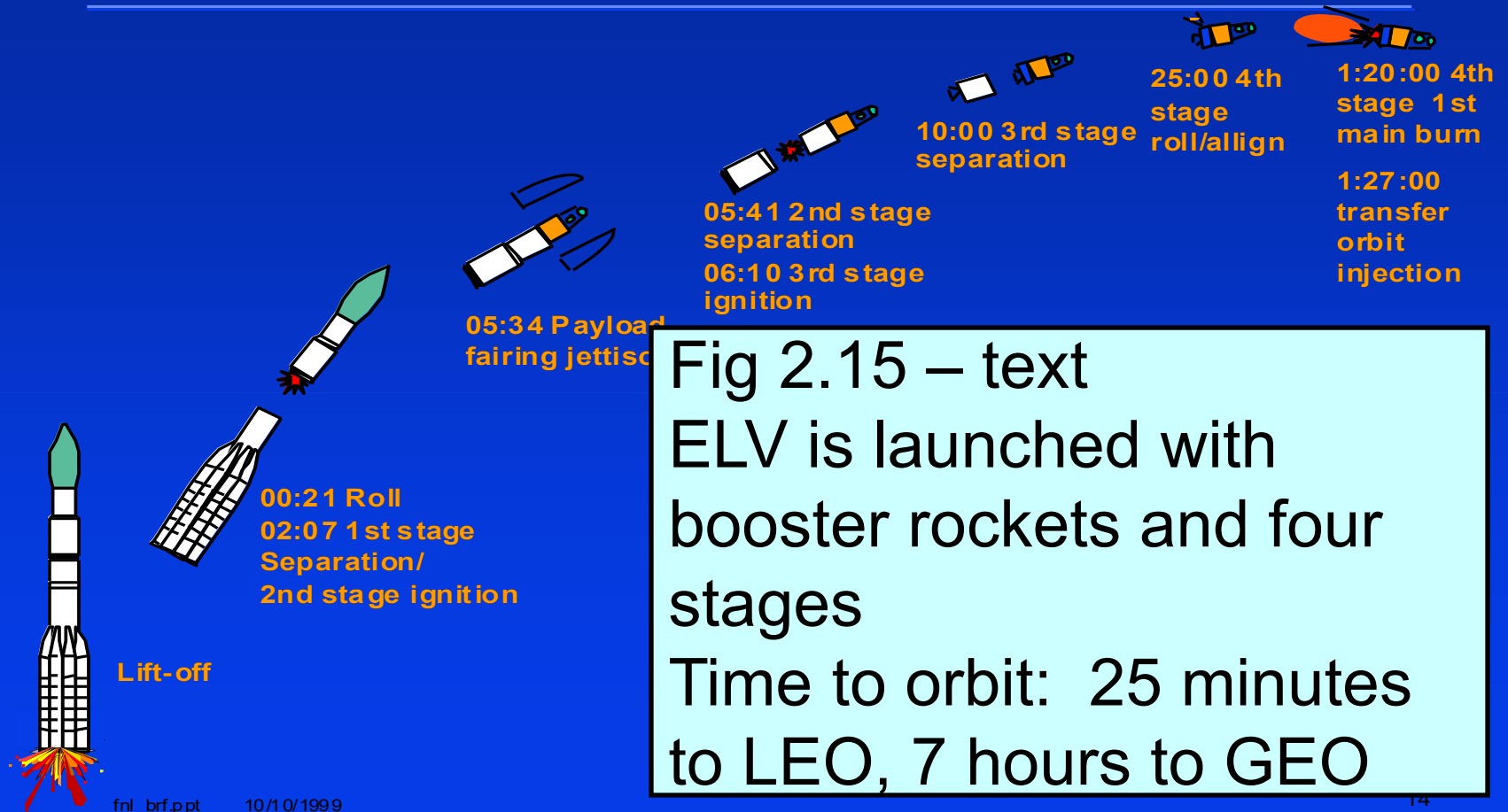
GEO arc as seen from northern hemisphere -
Sun passes through GEO arc in spring and fall

Launching Satellites

- Need to place satellite into an orbit with an appropriate combination of height and velocity
- Large quantities of fuel are generally required
- It is advantageous to shed excess weight as the launch progresses
- Leads to launching in multiple *stages*

Launching Satellites

Proton Launch Sequence



Launching Satellites

- With an *expendable launch vehicle* (ELV) the rocket components are not re-used (although they may be recovered)
- The space shuttle was an example of a *re-useable launch vehicle* (RLV)
- Commercial firms have developed RLVs to reach LEO (SpaceX, Blue Origin)
- SpaceX Falcon 9 boosters return and landed on platforms (if all goes well)



First stage booster of a SpaceX Falcon 9 launch lands on a drone ship in the Atlantic Ocean

Placing Satellites into GEO orbit

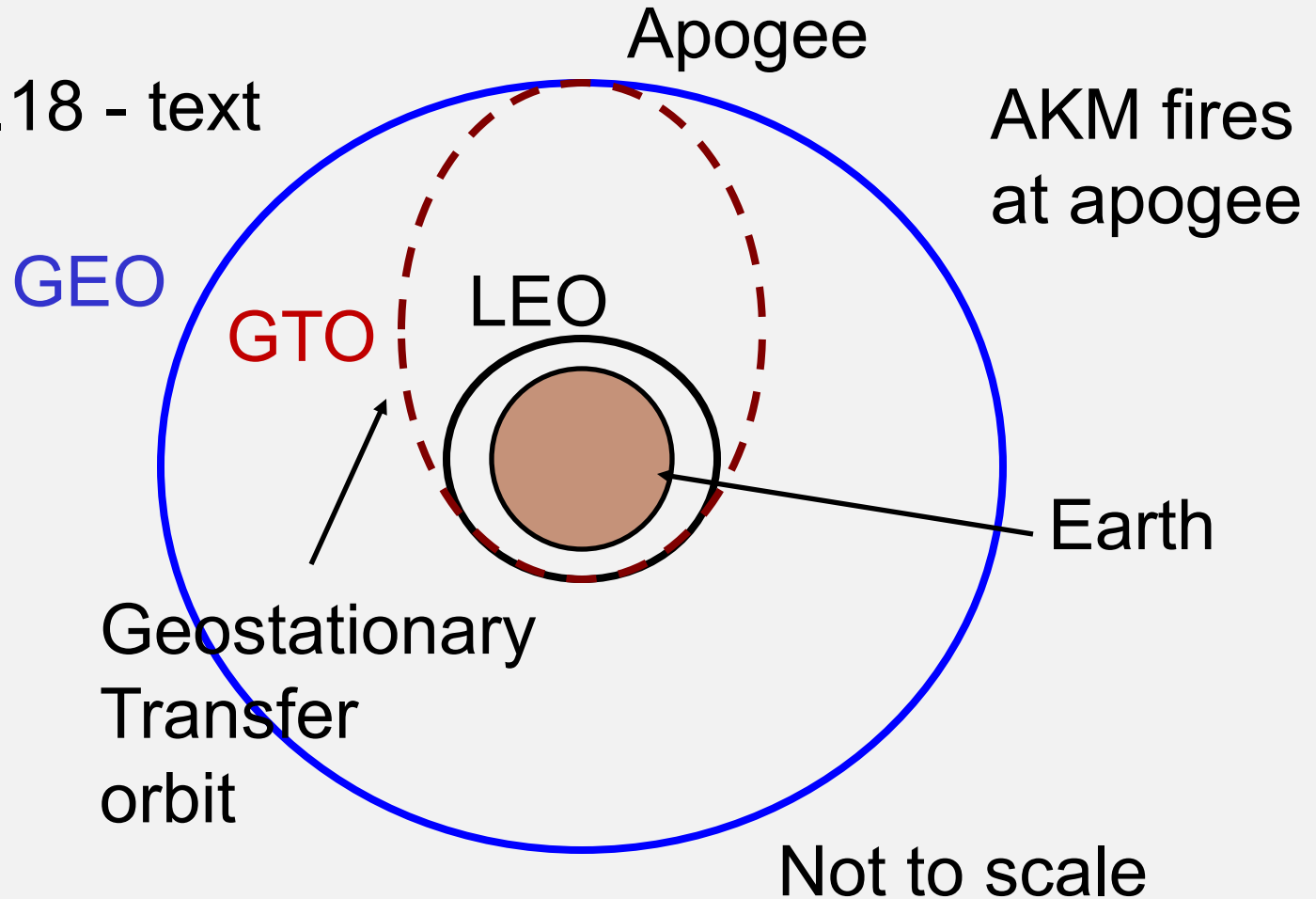
- With *direct insertion*, the final rocket stage lifts the s/c to GEO orbit
- Launch vehicle contains all the necessary propulsion energy to take the satellite into its correct orbit
- Satellite only needs to take care of in-orbit maneuvers , i.e., station-keeping, location changes, attitude corrections

Placing Satellites into GEO orbit

- Alternatively, GEO orbit can be reached in stages
- The launch vehicle places the s/c into LEO where the orbital elements are checked
- The final vehicle stage then inserts the s/c into *geostationary transfer orbit* (GTO)
- Finally, the s/c powers itself into GEO orbit

GTO / AKM Approach

- Fig. 2.18 - text



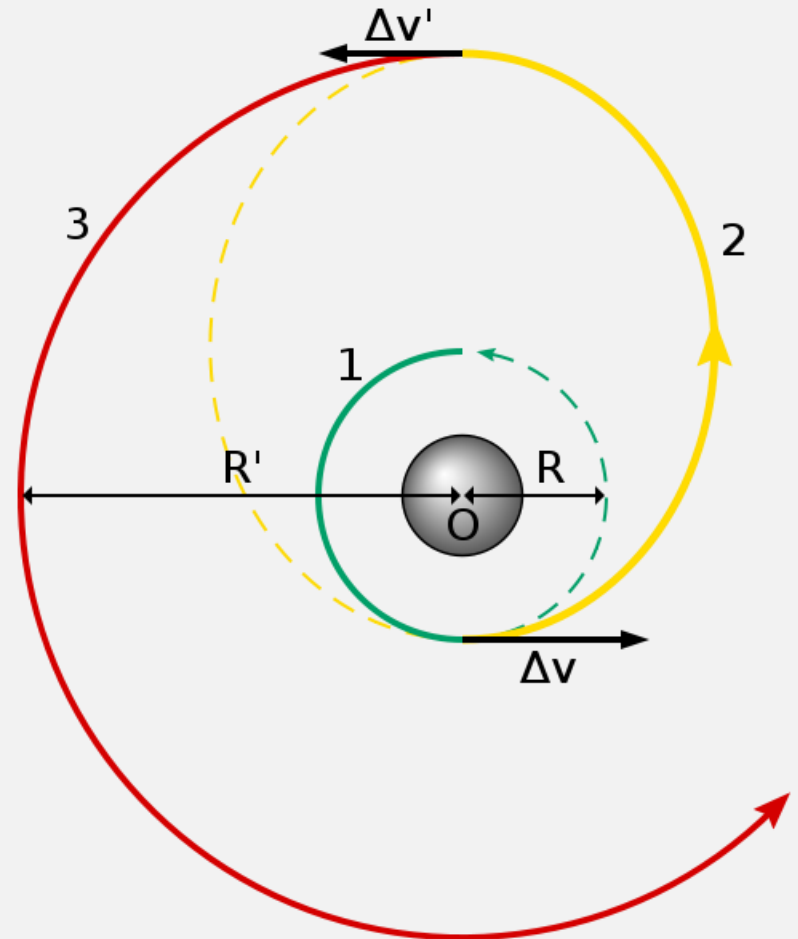
GTO / AKM Vehicle

- Launch to LEO
- Fire last stage of ELV (third or fourth) with satellite at perigee of LEO
- Satellite goes into elliptical *transfer orbit*
- Apogee of transfer orbit is at GEO altitude
- Fire *Apogee Kick Motor* (AKM)
- GTO is raised to a circular, geostationary orbit

Hohmann transfer orbit

- 1) Lower circular orbit
- 2) Elliptical transfer orbit
- 3) Higher circular orbit

Velocity increments indicate impulsive velocity changes due to engine firings



GTO / AKM Approach

- At apogee of transfer orbit:
 - fire AKM (normally integrated into the spacecraft)
- Velocity vector must also correct inclination to 0° (for GEO orbit)
- Initial velocity at apogee of GTO is low
- Increase to 3070 m/s to circularize orbit

GTO / AKM Approach (1/2)

- Cape Canaveral launch (28° N)
- Initial transfer orbit:
 - 150 km × 35,786 km altitude
 - 1585 m/s at apogee
- Launch eastwards to take advantage of earth's spin (prograde orbit)
- About 2% of take-off weight reaches GEO
- 6000 kg GEO satellite ... 300 ton ELV

GTO / AKM Approach (2/2)

- Modify transfer orbit:
 - add 1463 m/s to raise perigee to 35,786 km
- GEO insertion:
 - add 366 m/s to correct inclination to 0°
- Typical sequence duration with two rocket firings - 7 to 8 hours (quite fast)

Energy

- Why are rocket launch sites at low latitudes?
- Earth is spinning at 0.47 km/s at the equator
- Launch sites close to the equator impart this velocity to the rocket even before launch
- LEO velocity ~ 7.4 km/s
- Equatorial launch site has 6% velocity “boost”
- Approximately doubles the size of the potential payload

Energy

- Additional advantage of an equatorial launch – less (or zero) orbital inclination to remove
- Sea Launch was a company that provided launches from a platform towed to the equator (1999-2014)
- Best land-based launch site is Kourou in French Guinea (South America) - Latitude 5° N



Energy

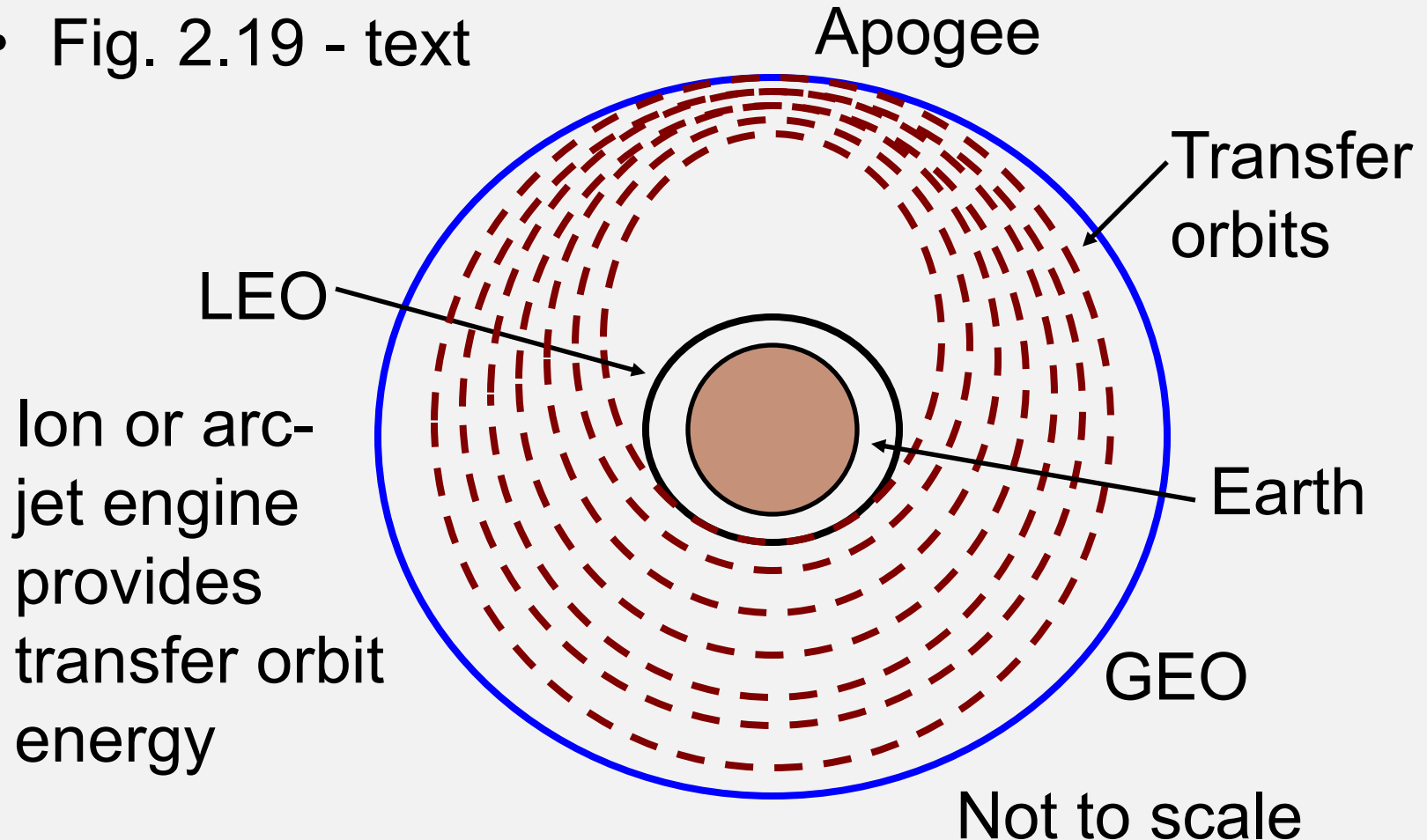
- US polar missions traditionally launch from Vandenberg AFB, CA (34.7°N)
- But in 2020 SpaceX started launching into polar orbit from Florida
- Polar orbit payload from Florida is 50% less than equatorial orbit payload

GTO / Slow Orbit Raising Approach

- As an alternative to an AKM, arc-jet and ion engines can provide small continuous thrust for a long time from solar panel power supply
- Satellite is deployed in GTO and continuous thrust engine is used to slowly increase velocity
- Eventually, orbit is circularized, in equatorial plane, at 42,164 km radius = GEO
- May take several months (rather slow)

GTO / Slow Orbit Raising Approach

- Fig. 2.19 - text



Slow Orbit Raising Advantages

- Avoids the shock to the satellite subsystems of an additional rocket firing (the AKM)
- Weight of fuel required by the AKM can be replaced by additional payload
- Reliance on electrical propulsion systems for station keeping can extend lifetime beyond that possible with fuel systems
- Consideration: satellite antennas and control systems need to be deployed in GTO in order to begin producing power

Deploying satellites for Megaconstellations

60 Starlink
satellites stacked
together before
deployment

Initial orbit is at
~290 km altitude



Once released the individual satellites use onboard electric propulsion (Krypton-fueled Hall thrusters) to reach their final orbits near 550 km altitude



A 'train' of Starlink satellites viewed just after release (Photo credit: explainingscience.org)

**** Breaking News** - Launch of Artemis II**

- The anticipated launch next month of the Artemis II mission will involve orbital stages
- The Space Launch System (SLS) core stage will place the Orion s/c into a highly elliptical transfer orbit with a low perigee
- The propulsion stage will then perform an apogee raise burn to achieve High Earth Orbit (HEO)
- In HEO the spacecraft will go beyond the GEO belt

NASA Schematic of Artemis II Trajectory



Heavy-Lift* Vehicles

*capable of lifting 22,000 kg to LEO

<u>Launch vehicle</u>	<u>Country</u>	<u>Payload</u>
• Delta IV Heavy	USA	28,790 kg
• Proton-M	Russia	22,776 kg
• Falcon 9 Full Thrust	USA	22,800 kg
• Long March 5	China	25,000 kg
• Ariane 5 (<i>medium lift</i>)	France	21,000 kg
• Vulcan Centaur	USA	27,200 kg

Super Heavy-Lift* Vehicles

*capable of lifting 40,000 kg to LEO

<u>Launch vehicle</u>	<u>Source</u>	<u>Payload</u>
• New Glenn	Blue Origin	70,000 kg
• SLS	NASA	130,000 kg
• <i>Long March 9</i>	<i>China</i>	<i>140,000 kg</i>
• Saturn 5	NASA	140,000 kg
• Falcon heavy	SpaceX	141,000 kg

- *Italics indicate first flight is pending*

First Falcon Heavy test
flight launch (February 6,
2018, Kennedy Space
Center)





Launch of Artemis I with the Orion spacecraft using the Space Launch System (SLS) rocket (November 16, 2022, Kennedy Space Center)

First flight of New Glenn (Blue Origin) rocket



Launch on January 16, 2025 from Kennedy Space Center

Developments in Launch Capabilities

- Satellites can also be launched from aircraft, e.g. Pegasus, Stratolaunch (halted January 2019), LauncherOne (1st launch, Jan. 17, 2021)
- Tourist launches into space are now taking place, e.g., New Shepard (Blue Origin), Virgin Galactic
- The drive to reduce costs is behind many of the technical developments
- A launch vehicle can be ‘shopped’ on the basis of price per kg to LEO or GTO (see Tables in text)

Manned Space Vehicles

- SpaceX – Cargo Dragon has carried supplies to the ISS since March 2019; Crew Dragon first successfully rotated crew on May 31, 2020
- Boeing Starliner – two uncrewed test flights (2021, 2022) encountered difficulties although the second did dock with the ISS; the first crewed flight launched on June 5, 2024 - it experienced leaks and the crew were left on the ISS. The craft returned safely.

Manned Space Vehicles

- Blue Origin – has made multiple suborbital flights for tourism with the New Shepard rocket. Reached orbit with the first launch of the New Glenn vehicle in January 2025
- Orion – the crew capsule for the Artemis program which is launched with the Space Launch System (SLS) rocket. Flew uncrewed to the moon with Artemis I. Will fly crewed with Artemis II

Design considerations

- Design lifetime
 - Electronics outlast station keeping fuel
 - GEO satellites must have E-W station keeping (stay in your slot!)
 - Some N-S drift can be tolerated
 - If N-S station keeping is ceased, orbit inclination increases at about 0.85° per year

Design criteria

- *EODL* End of design life
- Design criterion for payload (electronics)

~ 14 - 18 years

- *EOML* End of maneuvering life
- All station keeping fuel used up

~ 10 - 12 years

Design criteria

- Station keeping tolerance for GEO satellites is $\pm 0.05^\circ$
- Tolerance of $\pm 0.05^\circ$ needs little more fuel than ± 1.0 or $\pm 5^\circ$
- E - W and N - S station keeping maneuvers may happen at different times
- Fuel is used up in N - S maneuvers
- E - W corrections may be by ion engine

Summary (1/3)

- Satellite orbits are not perfectly Keplerian, station keeping is required
- Gravitational attractions of moon and sun pull satellite out of GEO orbit
- Atmospheric drag is the main perturbation to LEO orbits
- Expendable fuel makes up much of GEO satellite mass at launch, sets finite lifetime

Summary (2/3)

- Satellites must carry batteries for eclipse operation
- Satellites can be raised into orbits with rockets by direct insertion or by taking a staged approach with transfer orbits
- Slow orbit raising is achieved with electric propulsion systems
- The rotation of Earth gives a boost to launch velocity at lower latitudes

Summary (3/3)

- Access to space is now provided by commercial firms at much reduced cost
- Super-heavy lift launchers are coming on line to support the deployment of satellite mega-constellations and lunar missions



Liftoff of the Ariane 5 rocket from the Guiana Space Centre in Kourou, French Guiana at 5:20 p.m. EST, Thursday, Jan. 25, 2018.

Credits: Arianespace



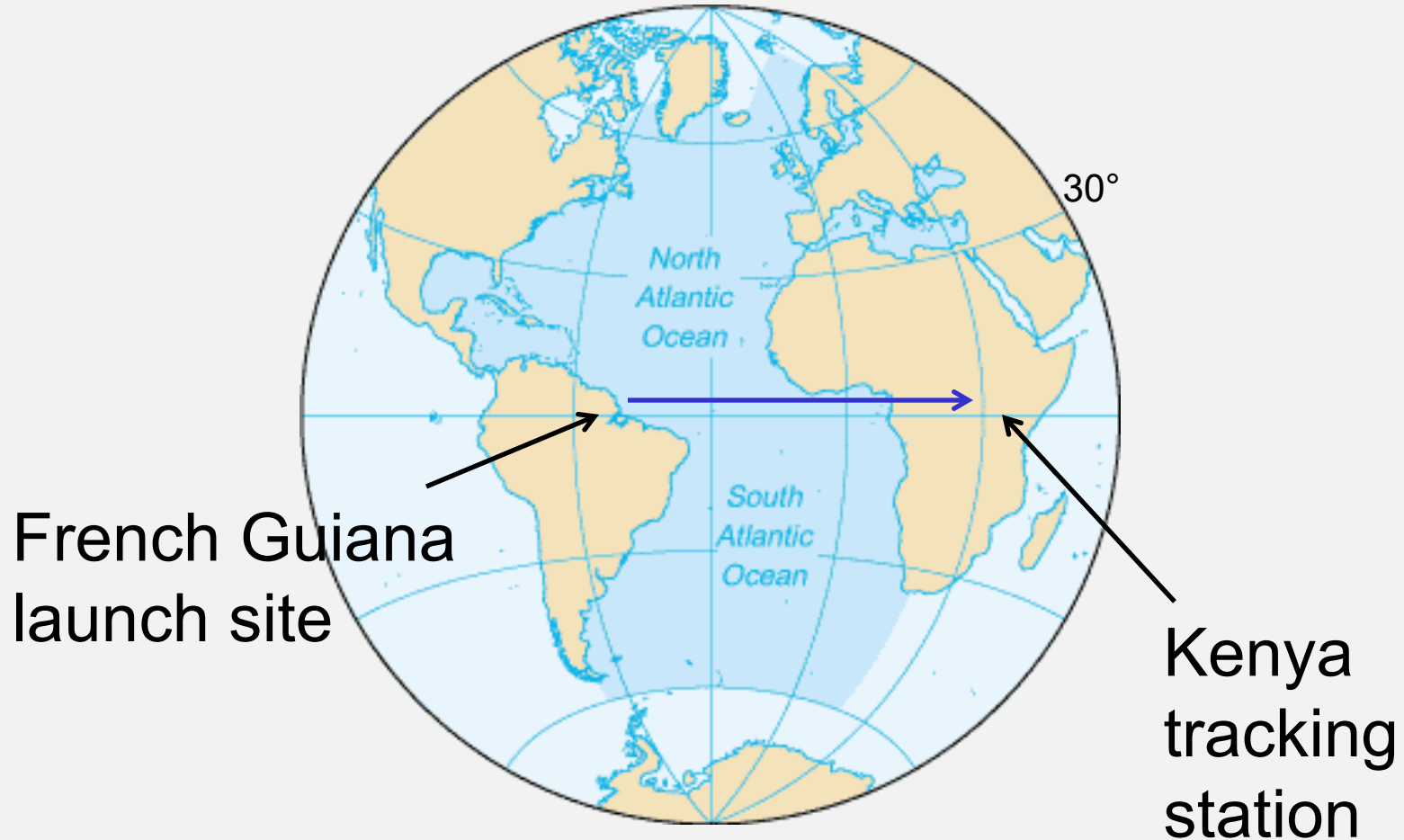
SES-14
communications
satellite at
launch facility in
French Guiana

The launch
vehicle was the
Ariane 5...a
launch anomaly
occurred

Launch from the Guiana Space Center

- Contact was lost 9 min 26 sec after liftoff (launch staff were reported to be 'pensive')
- Recovered after U.S. military detected a trajectory anomaly
- Intended inclination of GTO was 3° , wound up at 20.6°
- The Ariane 5 had strayed from its easterly course

Ariane 5 Launch of January 25, 2018 – intended track



Launch from the Guiana Space Center

- Ariane 5 launch of two large communication satellites intended for GEO orbit
- Perigee height = 232 km, Apogee height = 45,234 km (close to what was intended)
- The SES 14 telecom satellite used electric thrusters to reach GEO
- About one month of additional time was needed, for a total of $4 + 1 = 5$ months

SES-14 Communications Satellite

47.5°W



Ku-band coverage

Frequency Bands

- Best to learn the Microwave frequency bands:

L-band	1 – 2	GHz
S-band	2 – 4	GHz
C-band	4 – 8	GHz
X-band	8 – 12	GHz
Ku-band	12 – 18	GHz
K-band	18 – 27	GHz
Ka-band	27 – 40	GHz
V-band	40 – 75	GHz

Online resources for tracking satellites

- A number of online services provide tracking and viewing information
- One such commercial service is *In-The-Sky.org*
- Try this for link which will set your location to Blacksburg:

https://in-the-sky.org/satmap_worldmap.php?town=4747845#

- Especially try the '3D globe' tab and click on satellite symbols

Satellites as Spacecraft (s/c): Topics (Chapter 3)

- Communications satellites
- AOCS
- TTC&M
- Power system
- Communications system
- Antennas
- Reliability and Space Qualification

Communication Satellites

- Geostationary satellites have dominated communications until recently
- Cost of a large geostationary satellite on station: \$100 - \$500 M
- Non-Geostationary Satellite Orbit (NGSO) constellations in LEO are challenging this dominance
- Cost of an NGSO constellation: \$2 B

Satellite Systems

- Communications satellites require a number of systems to support their main mission
- Attitude and Orbit Control System (AOCS)
 - keeps satellite on station and antennas pointing in the correct direction
- Telemetry, Tracking, Command, and Monitoring (TTC&M)
 - allows controlling earth station to monitor satellite's health and send commands

Satellite Systems

- Thermal control system
 - keeps interior of satellite at correct temperature for operation of communication system
- Power system
 - generates electrical power from solar cells and provides correct voltages for equipment

Satellite Systems

- Communication system
 - revenue is earned by selling capacity
 - we will concentrate on this system
- Antennas
 - large satellites have complex communication antennas with shaped beams
 - there are separate antennas for the TTC&M systems

AOCS: Stability in Orbit

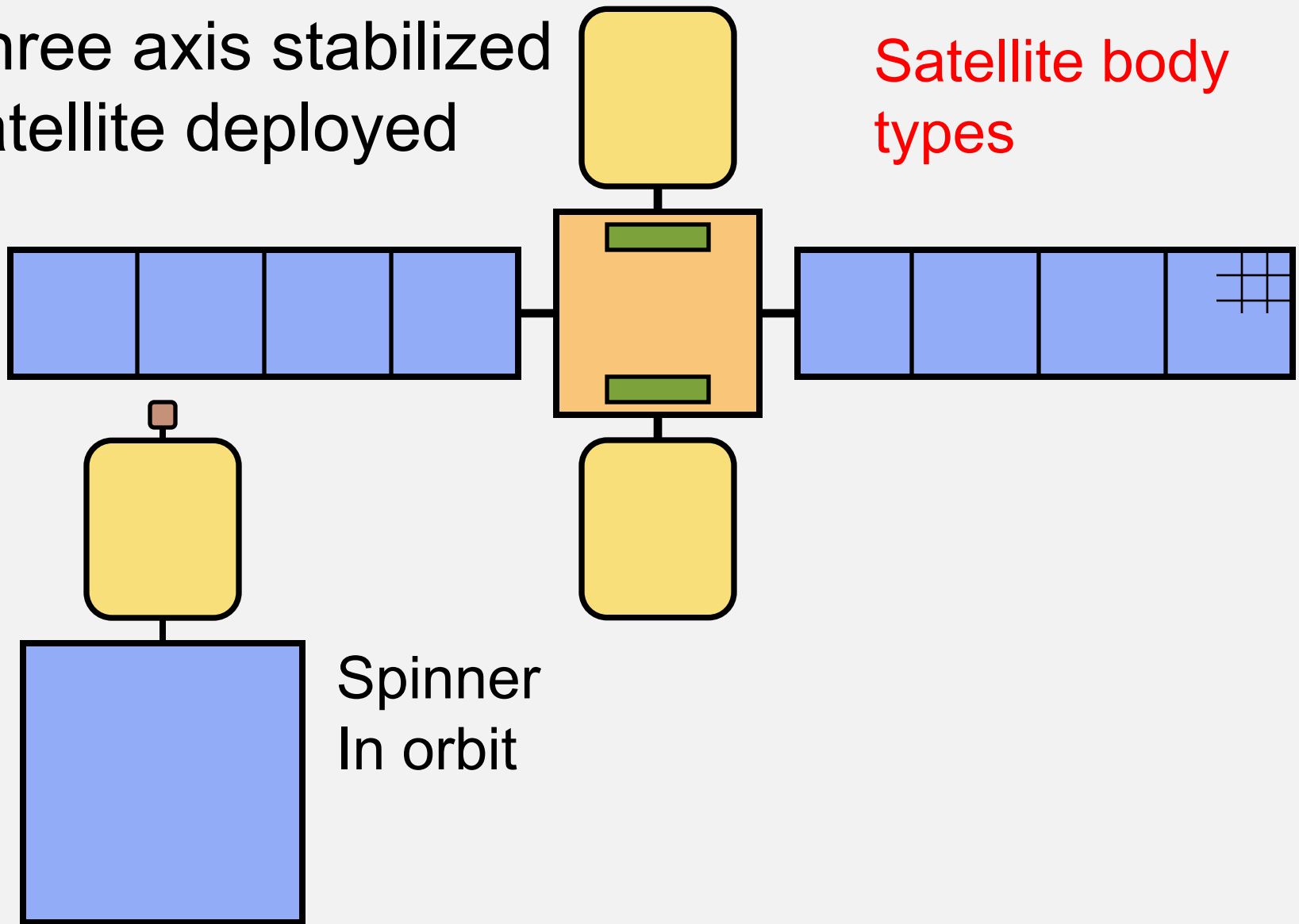
- Forces act on satellite that cause it to rotate and to drift out of position
- The satellite Attitude and Orbit Control System (AOCS) has two parts:
 - 1) attitude control – provides pointing stability
 - 2) orbit control – provides station keeping

AOCS - Attitude Control

- First consideration once on station: stabilize the satellite
- GEO satellites come in two designs
 - 1) Spin stabilized ('spinner'): Body of satellite rotates to provide gyroscopic stabilization
 - 2) Three axis stabilized: Has internal momentum wheel(s) to provide stability

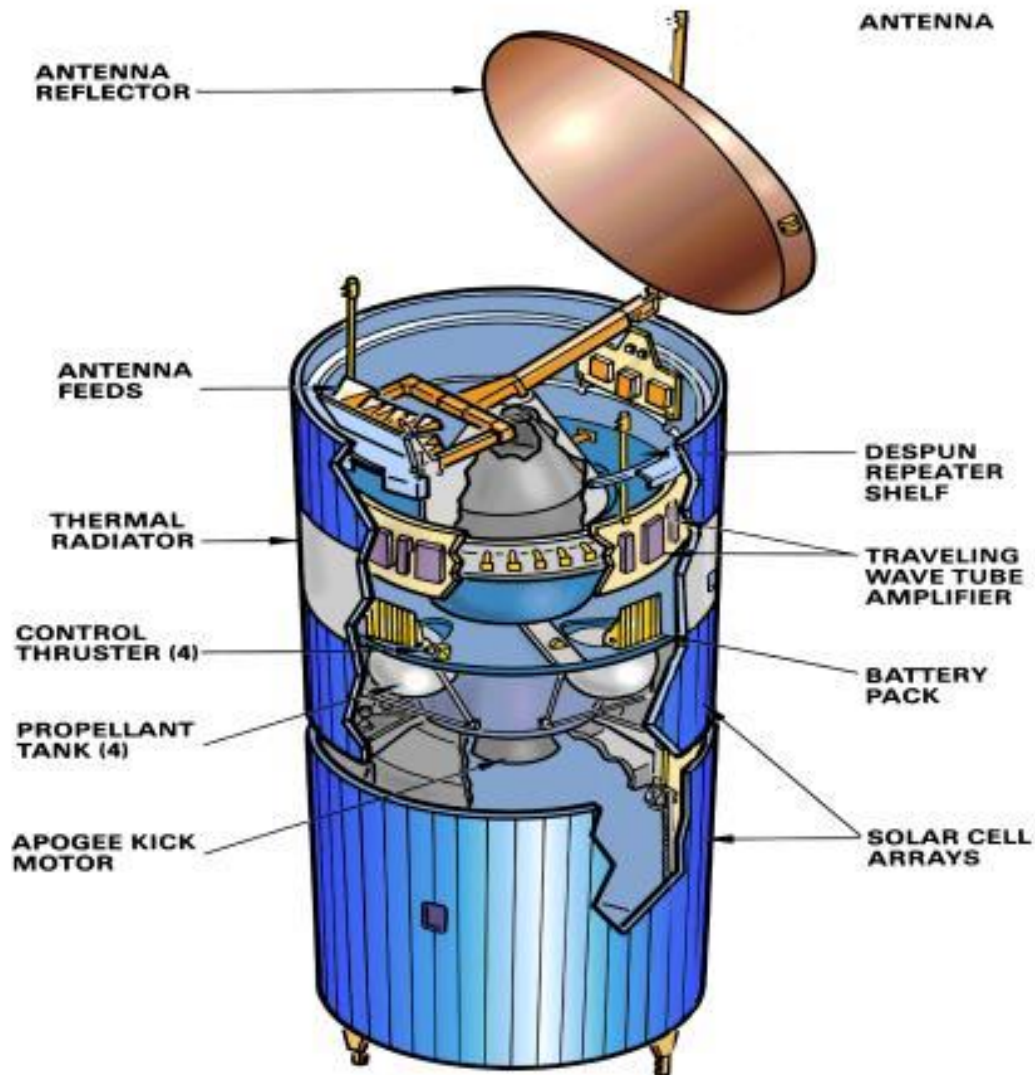
Three axis stabilized
satellite deployed

Satellite body
types



Spinner Satellite Design

- Key features:
- Solar cells coat large drum that spins
- Fuel tanks and AKM also spin
(AKM = Apogee Kick Motor)
- Gas jets on body used for attitude control
- Communications shelf and antenna *despun*
- Antenna must face earth at all times



Boeing 376 satellite – typical spinner

Spinner Satellite – Attitude Control

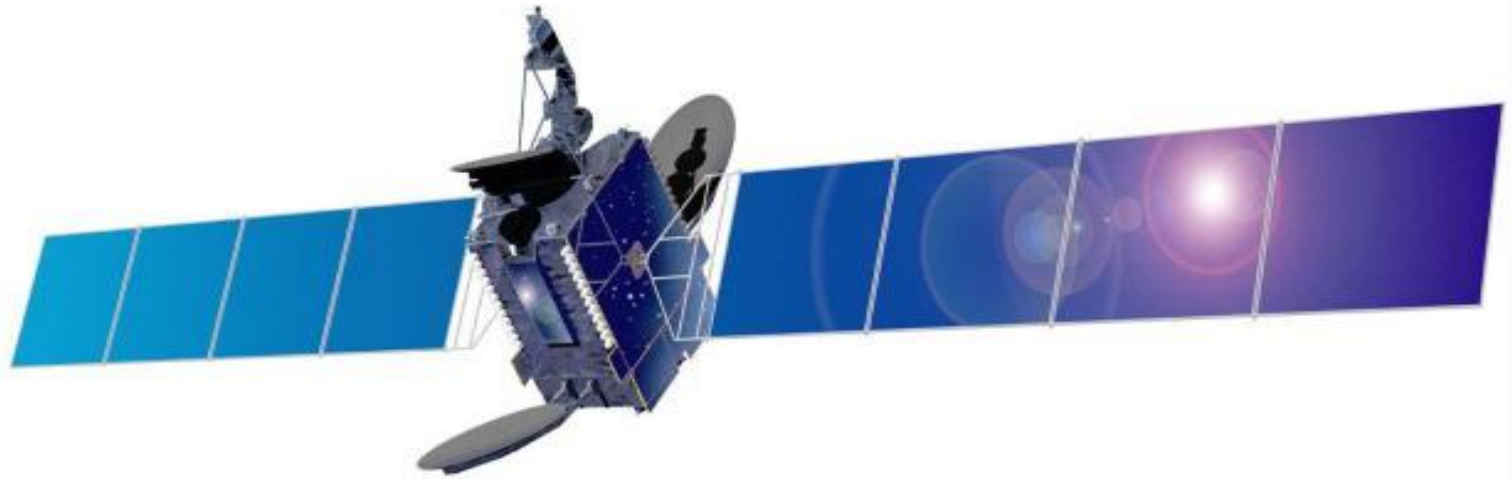
- Spinning most of the satellite provides powerful gyroscopic force
- Spin rate is typically 30 - 100 rpm
- Spin axis maintains its orientation perpendicular to orbit
- Forces will slowly change orientation of spin axis
- Gas jets must be fired to correct orientation

Spinner Satellite - Deployment

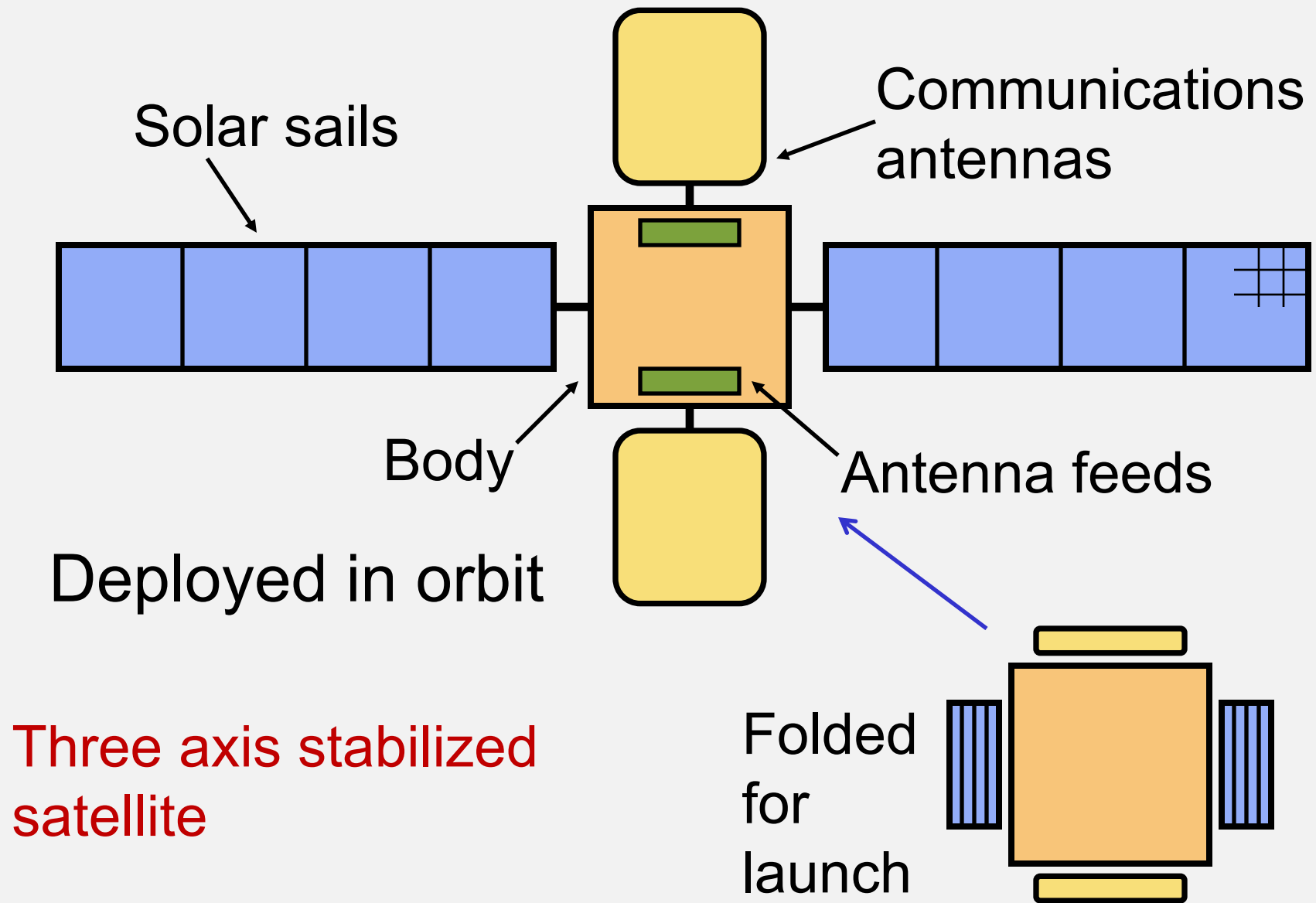
- The satellite is *spun up* using the radial gas jets
- Spinning satellite body requires antennas to be *despun* at same rate in opposite direction
- Requires electric motor and sensors
- Sensors look at infrared radiation from earth to determine and correct attitude

Three Axis Stabilized Satellites

- Key features:
- Solar *sails* or *wings* turn to face sun
- Fold up alongside body for launch
- Communications antennas fold down
- Body supports feeds for reflector antennas
- Body contains momentum wheel(s), AKM
- Fuel for AKM, gas jets



Echostar 5
Typical three axis stabilized satellite





Assembly of
SES-14
communications
satellite

Note transparent
shroud and folded
solar panels

Three Axis Stabilized Satellites

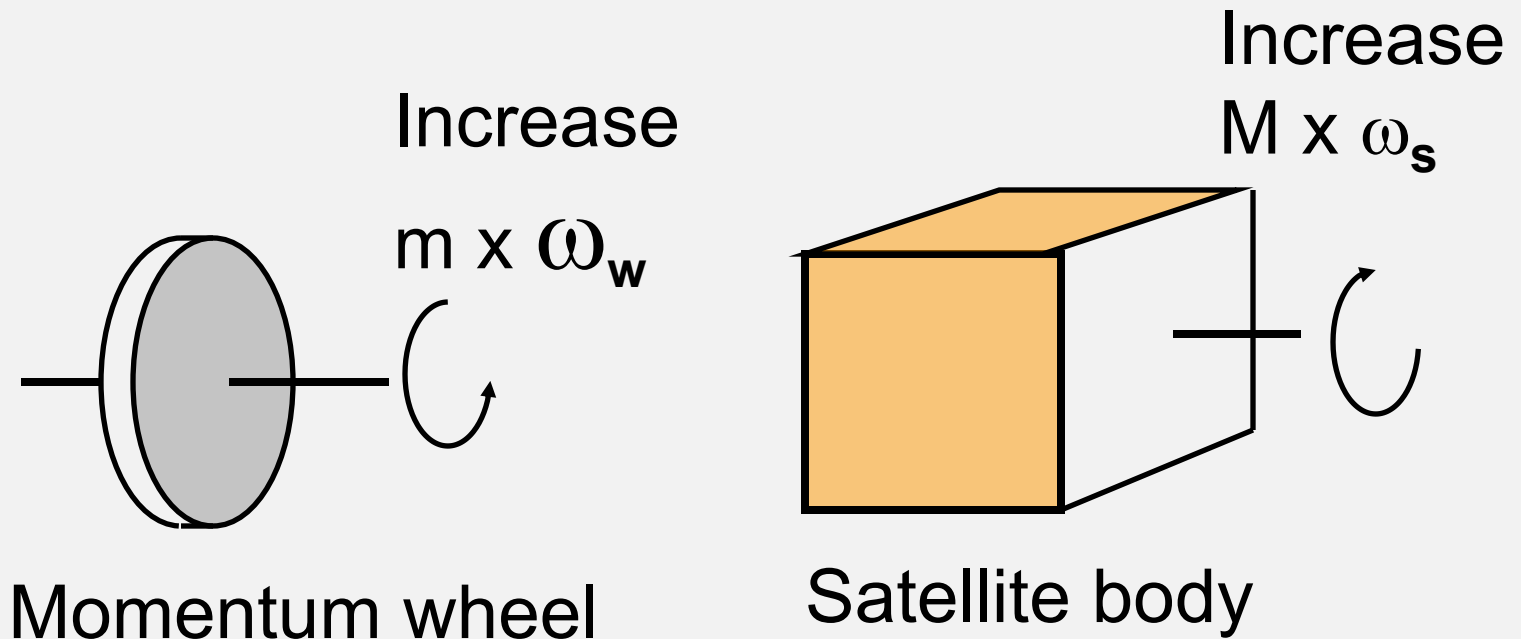
- Maximum dimensions:
 - Antenna diameter folded cannot exceed 3 m
 - Limited by diameter of ELV shroud
 - Satellite must be folded for launch
- When deployed in orbit:
 - Up to 30 m across its solar wings
 - 8 m across antennas and body

Three Axis Stabilized Satellite Attitude Control

- Angular momentum is given by mass $\times$ angular velocity = $m \times \omega$
- More precisely, $L = I \times \omega$, where I is the *moment of inertia*
- Moment of inertia takes account of how mass is distributed
- We will simplify and just talk about mass, m , in relation to angular momentum

Three Axis Stabilized Satellite Attitude Control

- Satellite has large M , small ω
- We can achieve same angular momentum with a heavy wheel rotating at high speed
- Small m , large Ω
- Conservation of angular momentum applies
- Undesired angular momentum of satellite body can be transferred to a *momentum wheel*



Increasing momentum of wheel will cause satellite body to rotate in opposite direction

Momentum wheels

- Satellite has large mass M , small ω_s
- Momentum wheel has small mass m , large ω_w
- $\delta (m \times \omega_w) = \Delta (M \times \omega_s)$
- Satellite can have three momentum wheels - one for each axis, or one momentum wheel on gimbals
- Speed is controlled from earth via TTC&M

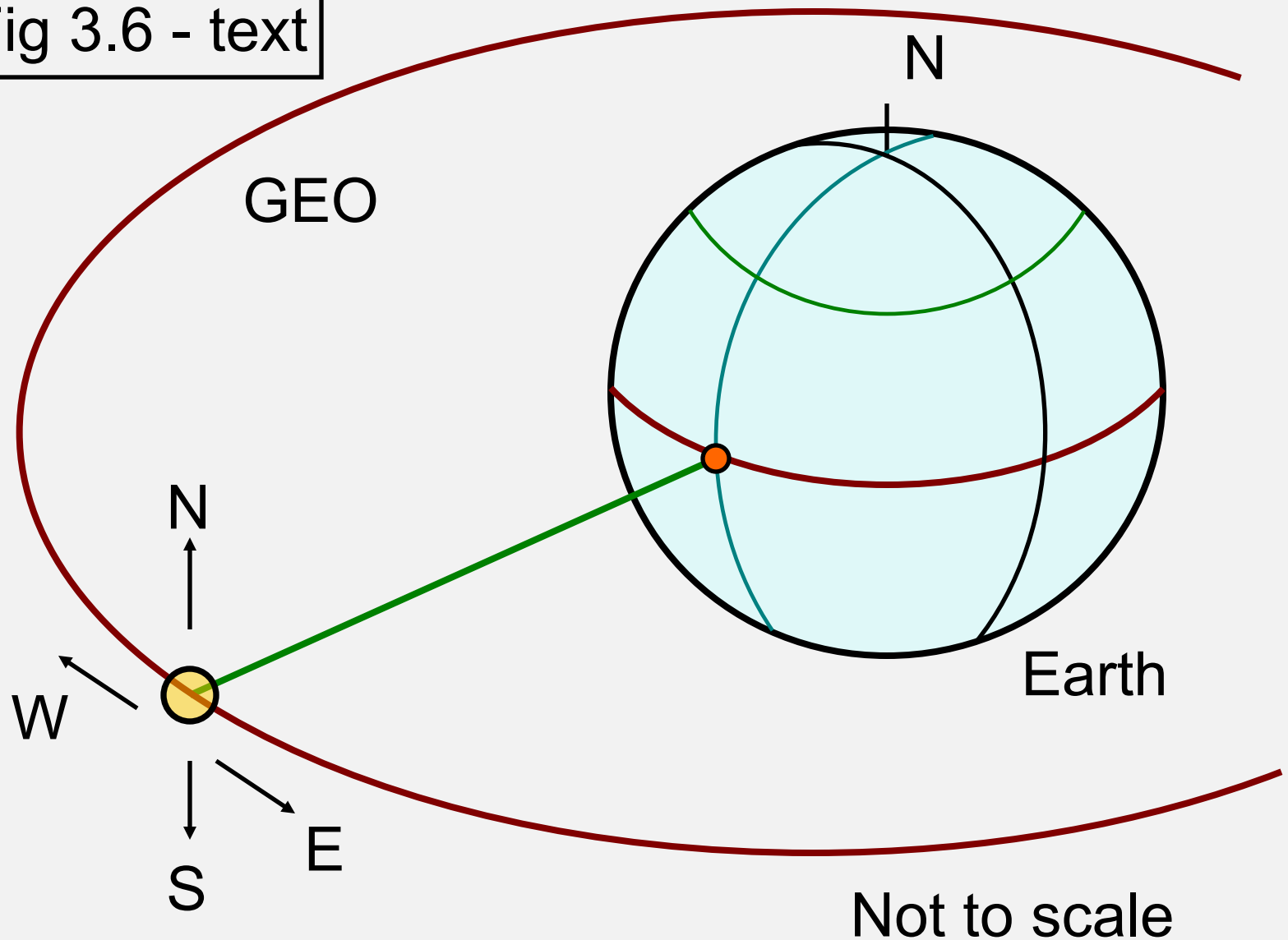
Communication Satellites

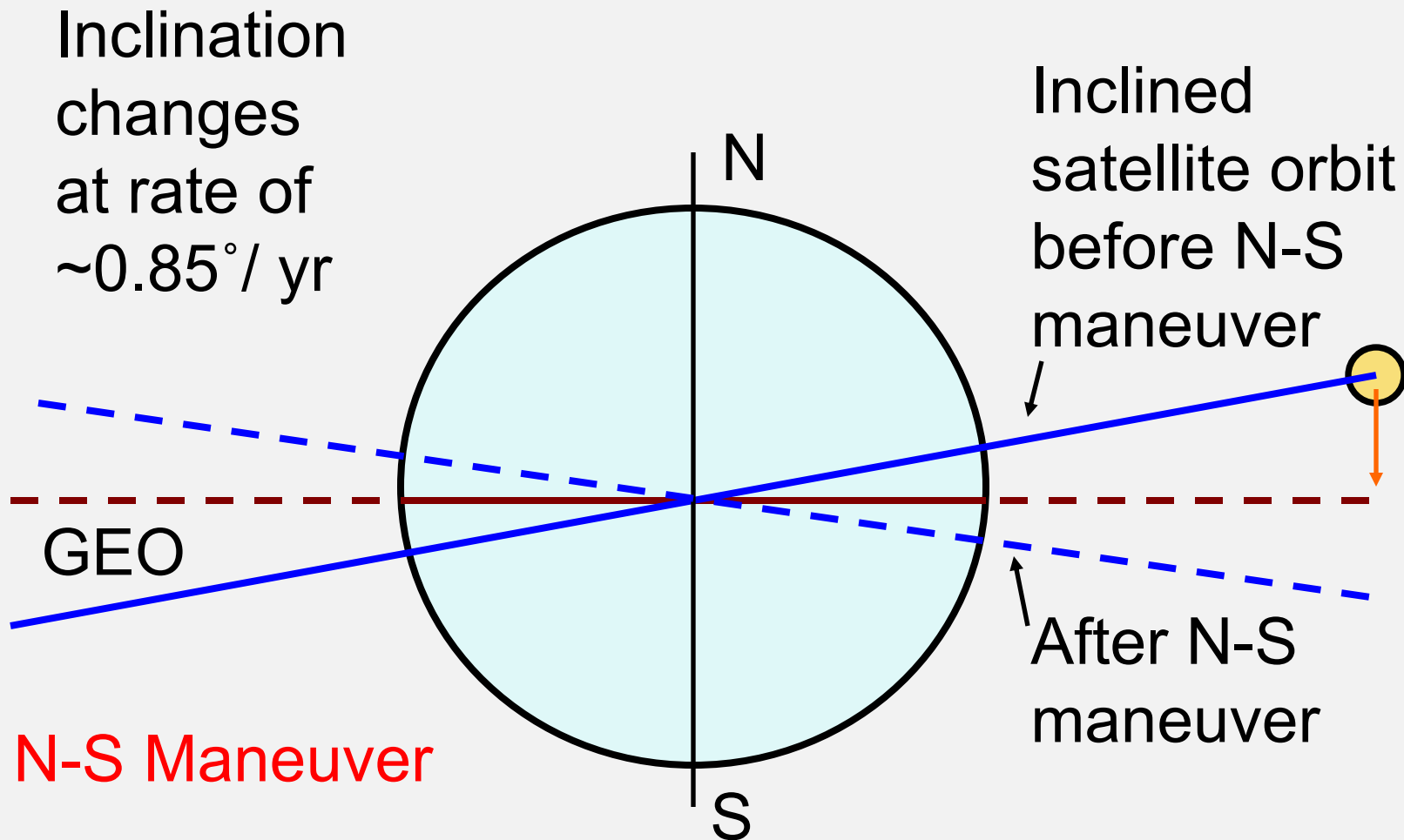
- Prior to 1990 most GEO satellites were spinners
- Now all large GEO satellites are three-axis stabilized, same for the NGSO constellations
- Spinners are generally limited to about 2 kW of electrical power
- Largest of three-axis designs generates 20 kW
- Thermal control was easier with spinners

AOCS - Orbit Control

- Non-spherical shape of Earth and sun / moon gravity cause satellite to drift out of orbit
- GEO requires: circular orbit, in earth's equatorial plane, correct velocity, and altitude
- Satellite drifts off station in E-W direction
- Inclination of orbit plane changes (N-S drift)
- Both effects must be countered with station keeping maneuvers

Fig 3.6 - text

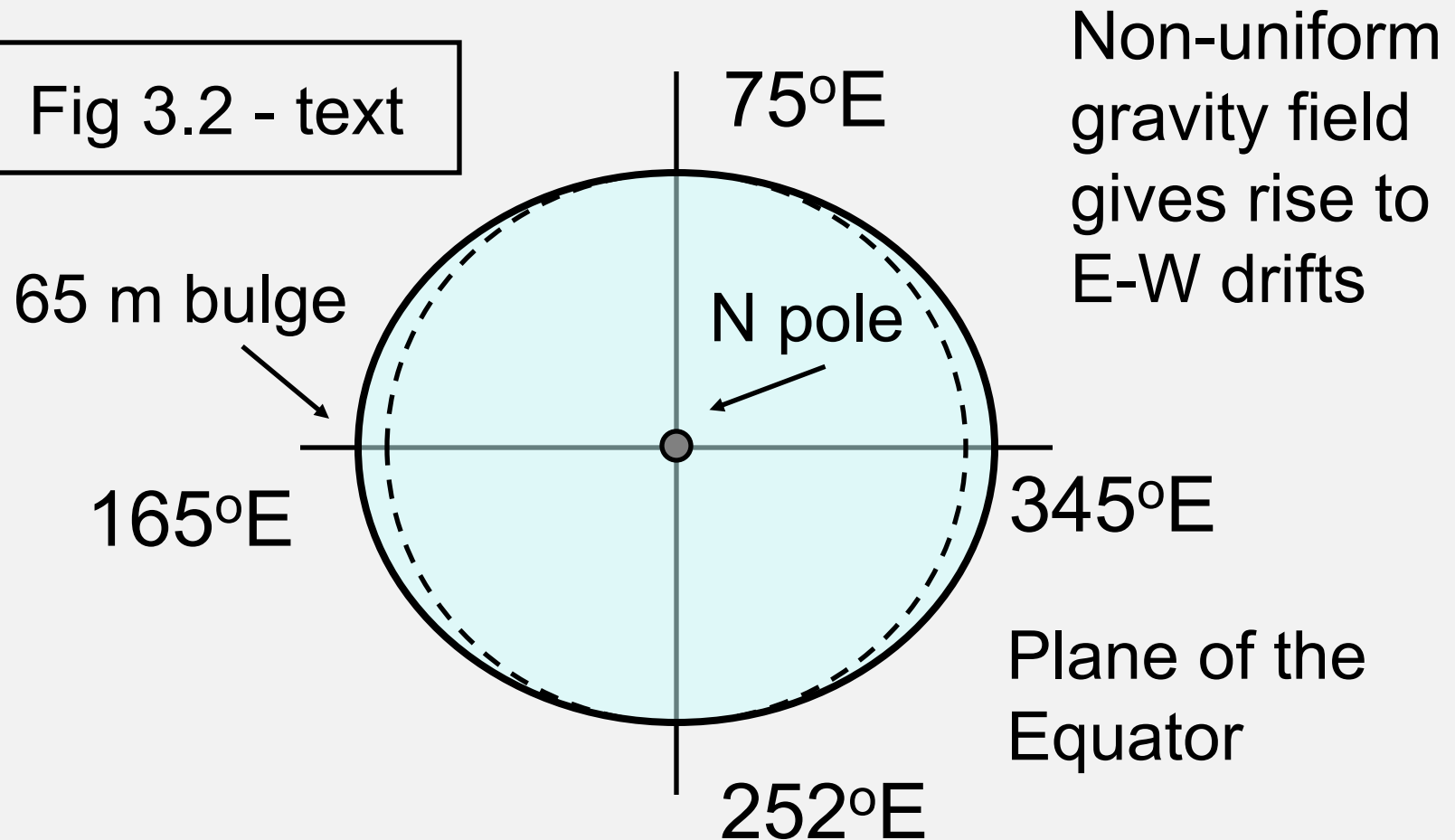




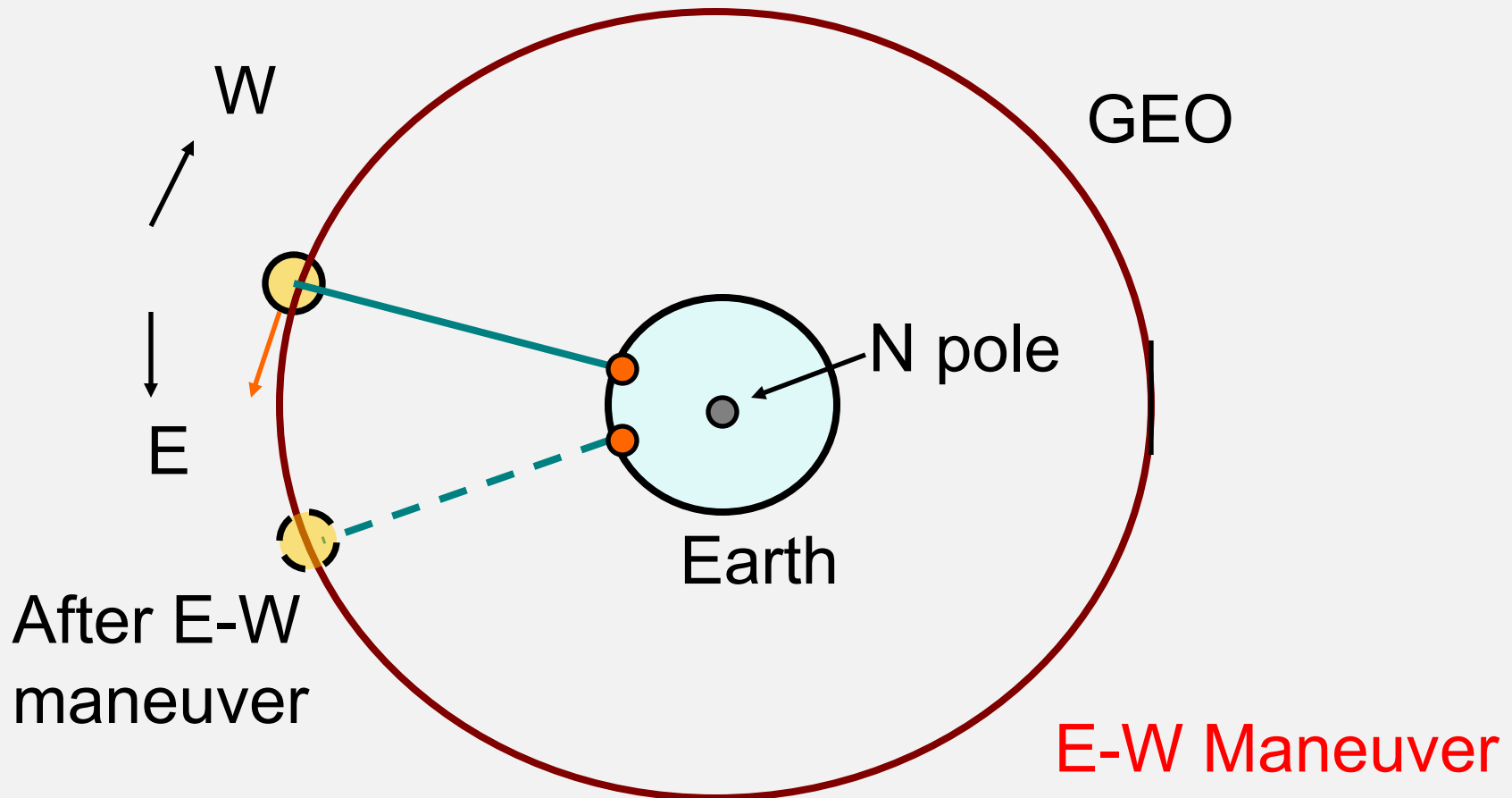
Satellite requires velocity vector normal to orbital plane, in N or S direction, to correct inclination

Not to scale

Fig 3.2 - text



- Stable points: 75° and 252° E
- Unstable points: 165° and 345° E



Satellite requires velocity increment in orbital plane to correct E-W drift

Not to scale

Notes on station keeping maneuvers (1/3)

- A maneuver consist of an initial thruster fire, a drift phase, and then an ending thruster fire
- Will typically pass through the nominal position during the drift phase
- The perturbing forces will then direct the satellite towards the nominal position

Notes on station keeping maneuvers (2/3)

- GEO satellites may relax their inclination to become *inclined orbit satellites*
- Such a satellite will be observed to drift about a mean position in the sky
- The ground antenna might need to track it or use a wider beam
- But a GEO satellites *must* maintain its E-W position (think Clarke belt)

Notes on station keeping maneuvers (3/3)

- A satellite in LEO orbit contends primarily with atmospheric drag - its orbit will decay
- Fuel is required to boost the satellite back to its target altitude (limits lifetime)
- The ISS requires a boost about once a month to maintain 370 – 460 km altitude (the ISS is refueled by propellant tankers)
- Atmospheric drag varies with solar activity (sunspots, flares, solar storms)

